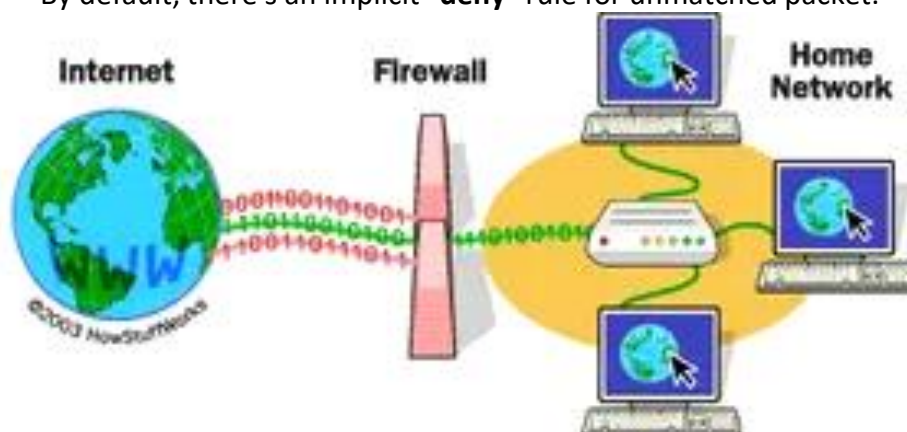


What is Firewall?

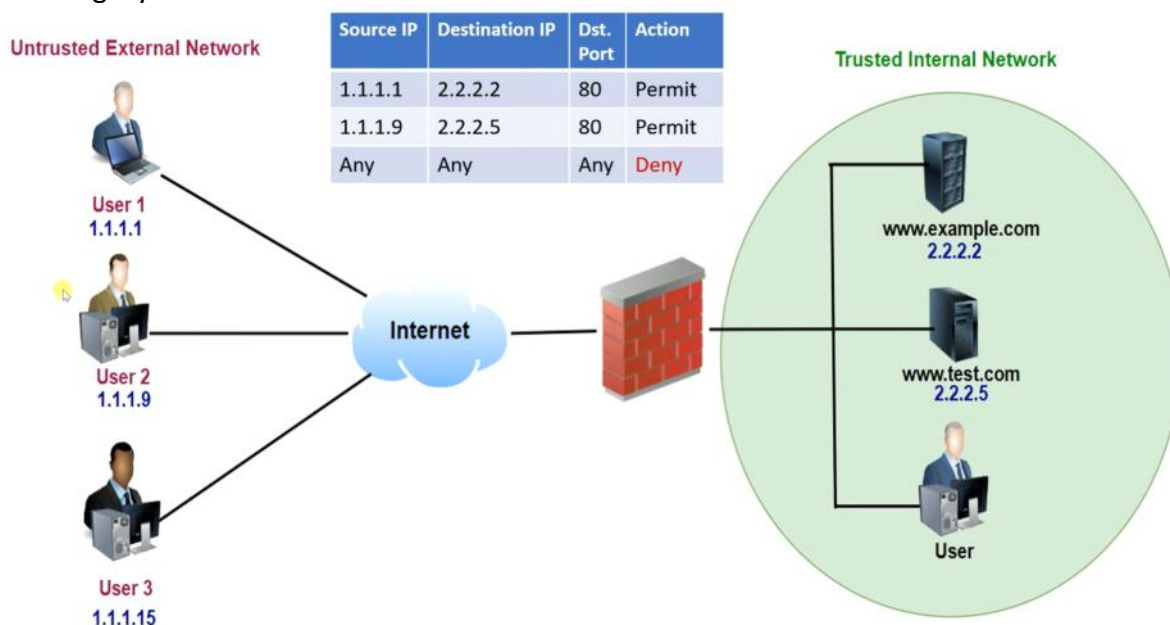
- It's the 1st line of defense.
- All incoming and outgoing network traffic will be passing only through it.
- It establishes a barrier between the trusted internal network(s) and the untrusted outside networks.
- It monitors all the incoming and outgoing network traffic that is passing through it and decides whether to permit or deny any specific traffic based on the set of security policy defined in it.
- By default, there's an implicit "**deny**" rule for unmatched packet.



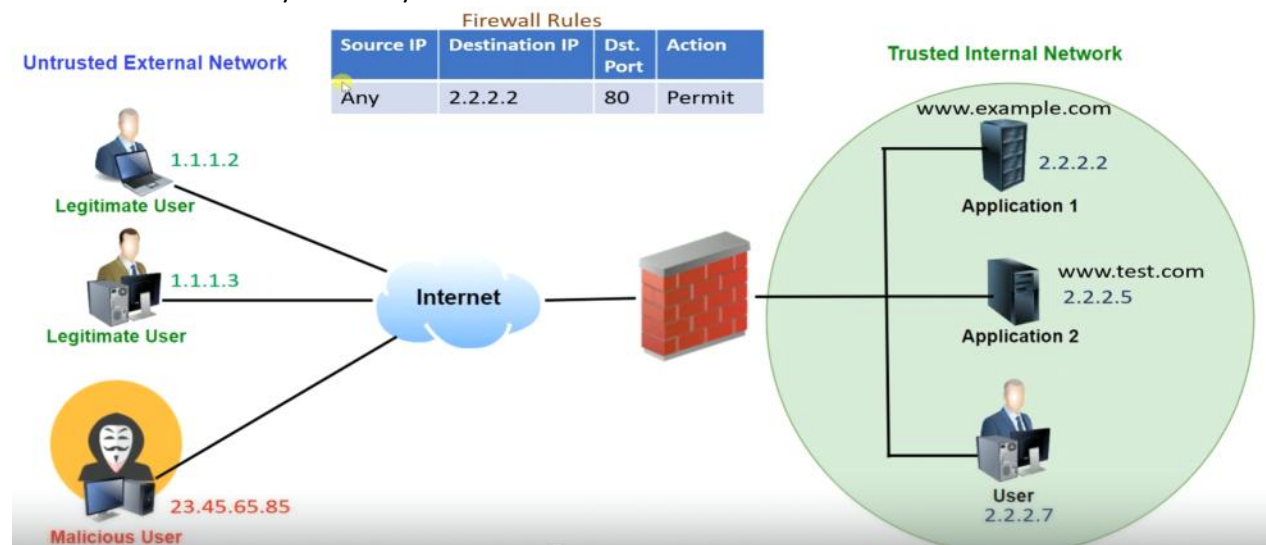
Simple legacy firewall rule:

Source IP	Destination IP	Dst. Port	Action
1.1.1.1	2.2.2.2	80	Permit
1.1.1.9	2.2.2.5	80	Permit
Any	Any	Any	Deny

Basic legacy network:



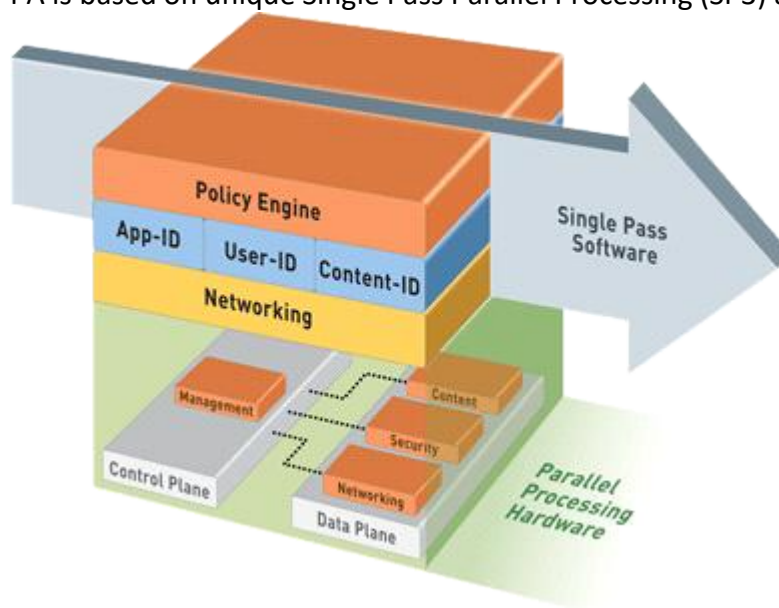
Note - Issue with legacy firewall is that it only works on L3 & L4 TCP/IP layers which checks only the source and the destination IP address. This could give access unauthorized access to the malicious user/attacker/hacker.



In this with legitimate users, Malicious user will also get the access to the www.example.com as source IP is allowed as "any" which gives the access to unwanted users too.

Palo-Alto Firewall

- Palo-Alto is a name of a city in California's San Francisco Bay Area in USA.
- PA was developed by Nir Zuk.
- Palo-Alto Networks (company) has a global footprint presence in 50+ countries, 24/7 support.
- It's a Next-Generation Firewall (NGFW) with advance features like:
 - Deep packet inspection
 - Intrusion detection and Intrusion Prevention system
 - Distributed Denial of Service (DDOS) protection.
 - Application-level inspection
 - User identification
 - Web content filtering,
 - Etc.
- It helps to identify, control and inspect the SSL encrypted traffic and applications.
- PA has a real-time content scanning feature to protect against viruses and malwares.
- It helps to protect against the spyware, data leakage and application vulnerabilities.
- It can help to prevent certain functions of an application without blocking the entire application.
- PA is based on unique Single Pass Parallel Processing (SP3) architecture.



- SP3 architecture provides high throughput and low latency network security service.
- SP3 has 2 components:
 - **Single Pass Software**
 - It performs the operation once per packet.
 - When a packet comes in, it performs all the operation at a same time in a single pass at once like:
 - networking function,
 - policy lookup,
 - application identification,
 - signature validation against malware or threats
 - content identification,
 - user identification and so on.
 - **Parallel Processing Hardware**

- The parallel processing hardware ensures the single pass software works smoothly and efficiently.
 - It has separate control plane and data plane with multiple cores and processors for smooth and better performance.
- 3 main advance features of PA firewall:
 - **App-ID** - identify the application
 - **Content-ID** - Scan the content
 - **User-ID** - Identify the user

Palo-Alto VM using Azure

Thursday, December 25, 2025 8:19 PM

Minimum requirements:

- **16 GB RAM**
- **4 vCPUs**
- **224 GB HDD/SSD**
- VM Image: Palo-Alto (**Panorama**)
- Username: <given at the time of VM creation>
- Password: <given at the time of VM creation>
- IP Address: <given at the time of VM creation>
- URL to access PA mgmt portal: <https://<IP-Address>>

Deployment using Azure ([link](#)):

1. Palo Alto Networks Professional Service Flex Licensing Migration Lab

1.1 Overview

The Following Lab guide will help you to understand how to migrate Non-Flex licensed Software Firewalls from Non-Flex license model (ELA, etc) to the new Flex License Model. It will also cover how to create Deployment profiles in the Customer Support Portal (CSP) to cover several scenarios. The Lab will only cover Migration use cases as listed below

1. Standalone Firewall with Access to the CSP
2. Standalone Firewall, No Access to the CSP
3. Panorama-Managed Firewalls with Access to the CSP
4. Panorama-Managed Firewalls, No Access to the CSP

Private Cloud and other Public Cloud Providers will not be covered in the Lab. The deployed firewall are running in PanOS 9.1, 10.0.3, 10.2

1.2 Covered Scenarios

The following Scenarios and Lab activities are covered

1. Deploy a new Lab Panorama to fulfill the Migration process
2. Configure Panorama to perform the Lab activities
3. Setup the Customer support Portal (CSP)
 1. Creating several Deployment Profiles
4. Deploy Software Firewalls and License them with an ELA License
 - 1.2 Firewalls in PanOS 9.1.13-h3
 - 2.2 Firewalls in PanOS 10.0.9
 - 3.2 Firewalls in PanOS 10.2.3
5. Migrate Software Firewalls from NON-Flex License Model to Flex-License Model
 1. NON-Flex to Flex-License (Fixed Deployment Profile)
 2. Flex-License to Flex-License (Flexible Deployment Profile)
6. How to update the Deployment Profile
 1. Enable/Disable CDSS
 2. Increase/Decrease vCPU count
7. Troubleshooting

2 Deploy Panorama in Microsoft Azure

For this workshop you will create a first a Panorama before we deploy the Software Firewall in other Public Cloud Providers. The Panorama will have direct internet access. the Panorama is not connected to any other internal Resource in Azure.

2.1 Deploy Azure Resource Group

1. Login in to Azure Portal (<https://portal.azure.com>). As Login use your Palo Alto Networks Credentials



2. Open Azure Cloud Shell



3. In Cloud Shell execute the following command but change before the values [StudentRGName] and [Location] Available Regions are: North Europe, East US, UK South, UAE North, Australia Central
- ```
az group create --name [StudentRGName] --location [Location] --tags Owner=Workshop-DeleteMe
```
4. The Output should look like the following

```
PowerShell PS /home/torsten> az group create --name InstructorRG --location northeurope
{
 "id": "/subscriptions/d471af8-9795-4e86-bbce-da72cfd0f8ec/resourceGroups/InstructorRG",
 "location": "northeurope",
 "managedBy": null,
 "name": "InstructorRG",
 "properties": {
 "provisioningState": "Succeeded"
 },
 "tags": null,
 "type": "Microsoft.Resources/resourceGroups"
}
```

#### 2.2 Deploy Panorama in Azure

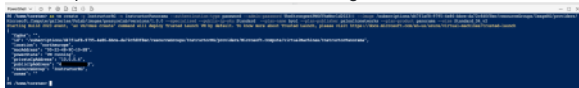
As next we will create the Panorama from a pre-staged image, after successfully creating the Resource Group.

1. Please go back to the Azure Cloud Shell
2. In the following command update the following variables with yours:
  1. [StudentRGName] Use the same name of the previous created Resource Group in the Chapter [Deploy new Resource Group in Azure](#)
  2. [VM-Name]
  3. [YourPassword]

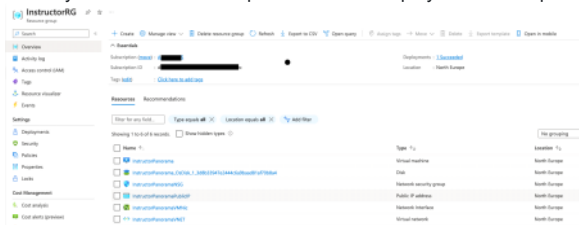
Don't change any other variables

```
az vm create -g [StudentRGName] -n [VM-Name] --authentication-type password --admin-password [YourPassword] --image /subscriptions/d47f1af8-9795-4e86-bbce-da72cfd0f8ec/resourceGroups/ImageRG/providers/Microsoft.Compute/galleries/PsLab/images/psazurelab/versions/1.0.0 --specialized --public-ip-sku Standard --plan-name byol --plan-publisher paloaltonetworks --plan-product panorama --size Standard_D4_v2
```

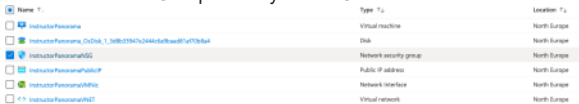
1. After you made the changes, execute the command in Azure Cloud Shell
2. The Output should look like the following



3. Check your Resource Group in Azure if the Deployment is completed



4. In the Resource Group select your NSG



5. Now create an Inbound Security Rule to allow any traffic to your newly created Panorama



6. Login to your Panorama via the Public IP associated to it. The Instructor will provide you the Username and Password.

1. [https://\[Public-IP\]](https://[Public-IP])



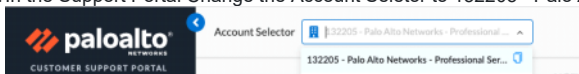
Congratulations!! You have successfully deployed a Panorama in Microsoft Azure.

### 3. Customer Support Portal

In the following Lab section we will go to the Customer support portal (CSP) to create your first Deployment Profiles. This is needed for the initial Migration and generating a Serialnumber for the Panorama

#### 3.1 Login To Customer Support Portal

1. Login with your PANW Credentials at the Customer Support Portal <https://support.paloaltonetworks.com/>
2. In the Support Portal Change the Account Selector to 132205 - Palo Alto Networks - Professional Services



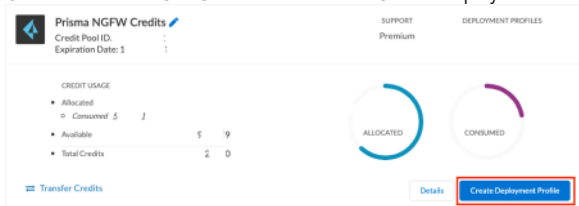
3. On the Support Portal Page on the left side go to Assets -> Software NGFW Credits

#### 3.2 Create Fixed Deployment Profiles

Now you will create one Deployment Profile in the Customer Support Portal.

##### 3.2.1 Azure Deployment Profile

1. On the Prisma NGFW Credits Pool click on Create Deployment Profile



2. Select the following as shown on the picture below and click Next

## Create Deployment Profile

\* Select firewall type:

☒ VM-Series

☐ CN-Series

\* Select a vCPU configuration type:

☒ Fixed vCPU models (Valid for all currently supported PAN-OS releases)

☐ Flexible vCPUs (PAN-OS 10.0.4 and above)

Cancel Next

3. In the Deployment Profile use the following and replace Instructor-Lab under "Profile Name" with "Migration-Lab-Fixed-[StudentName]"

## Create Deployment Profile

VM-Series

Profile Name: Migration-Lab-Fixed-Instructor

\* Number of Firewalls: 6

\* Fixed vCPU model: VM-300 (4 vCPUs)

\* Security Use Case: Custom

Customize Subscriptions

☒ Advanced URL Filtering

☒ DNS

☐ Global Protect

☐ DLP

☐ SD-WAN

☐ Intelligent Traffic Offload

☒ Advanced Threat Prevention

☐ Hyperscale Security Fabric (Beta)

☐ Web Proxy (Promotional Offer)

☐ Advanced Wildfire

> Legacy Subscriptions

Use Credits to Enable VM Panorama

☒ For Management

☐ As Dedicated Log Collector

Protect more, save more

Calculate Estimated Cost

Cancel Create Deployment Profile

4. Click "Create Deployment Profile"

5. Verify that your Deployment Profile is successfully created

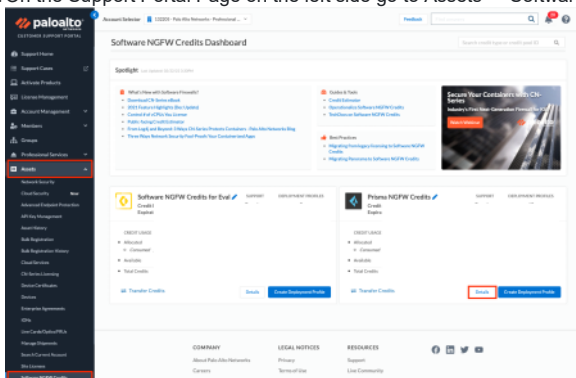
| Profile Name   | Firewall Type | Fixed vCPU models (Valid for all currently supported PAN-OS releases) | Number of Firewalls | Fixed vCPU model | Security Use Case | Advanced URL Filtering              | DNS                                 | Global Protect           | DLP                      | SD-WAN                   | Intelligent Traffic Offload | Advanced Threat Prevention          | Hyperscale Security Fabric (Beta) | Web Proxy (Promotional Offer) | Advanced Wildfire        |
|----------------|---------------|-----------------------------------------------------------------------|---------------------|------------------|-------------------|-------------------------------------|-------------------------------------|--------------------------|--------------------------|--------------------------|-----------------------------|-------------------------------------|-----------------------------------|-------------------------------|--------------------------|
| Instructor-Lab | VM            | Fixed vCPU models (Valid for all currently supported PAN-OS releases) | 6                   | VM-300 (4 vCPUs) | Custom            | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/>    | <input checked="" type="checkbox"/> | <input type="checkbox"/>          | <input type="checkbox"/>      | <input type="checkbox"/> |

### 3.3 License Panorama

In the next steps you will create a Serialnumber for your previous created Panorama with the Flex License Credits

#### 3.3.1 Provision Panorama Serialnumber

1. Login with your PANW Credentials at the Customer Support Portal <https://support.paloaltonetworks.com/>
2. On the Support Portal Page on the left side go to Assets -> Software NGFW Credits -> Details

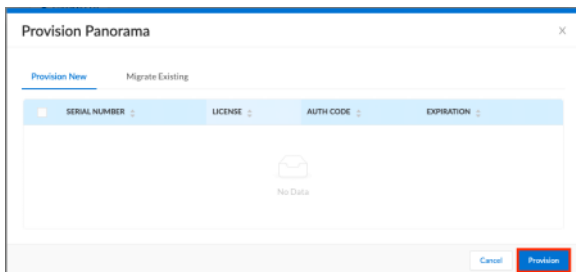


3. Now Search for your previous created Azure Deployment Profile [here](#)

4. Click on the 3 dots and on Provision Panorama



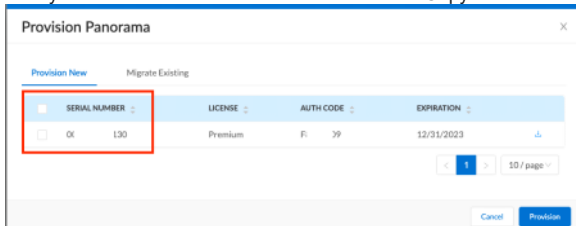
5. In the new window click on Provision



6. Once the window is closed repeat the steps from step 3



7. Now you can see a Serialnumber in the Window. Copy and Paste the Serialnumber



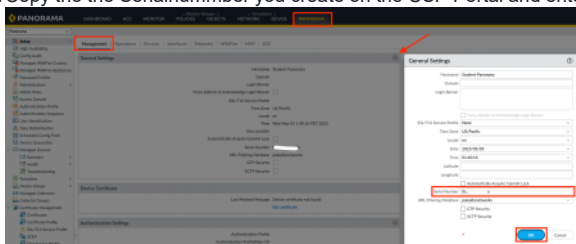
8. You can close the window by clicking Cancel

### 3.3.2 Configure Panorama

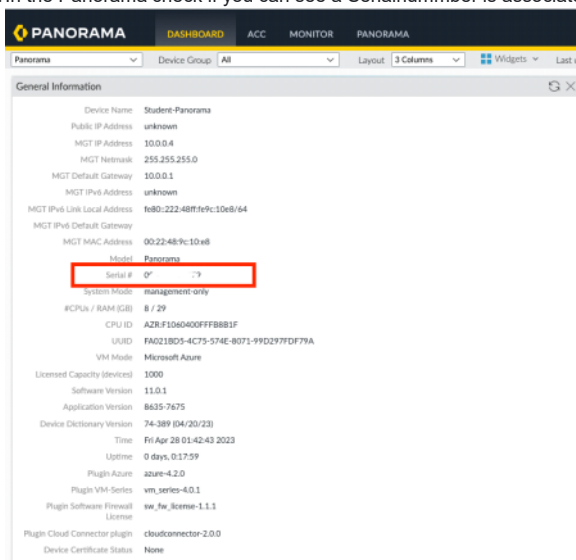
As next we will License your Panorama with the Serialnumber you created above and create a new Decive Group and Template inside your new Panorama and do some basic configuration in your Device Template

#### 3.3.2.1 License Panorama

1. Login to your Panorama [https://\[Public-IP\]](https://[Public-IP])
2. Copy the the Serialnummber you create on the CSP Portal and enter it under the Panorama Tab -> Setup -> Management -> General Settings

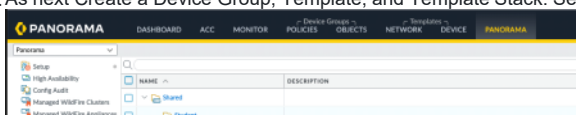


3. Hit OK and reload the UI. Check if a pending commit on the Panorama is needed. If yes, commit to Panorama.
4. In the Panorama check if you can see a Serialnummber is associated to it

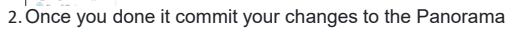


#### 3.3.2.2 Create Device Group and Device Template

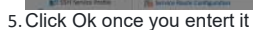
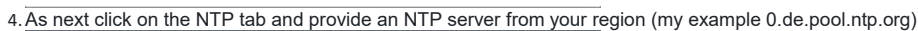
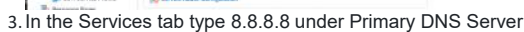
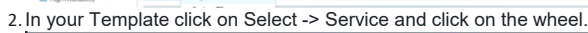
1. As next Create a Device Group, Template, and Template Stack. See the picture below as example





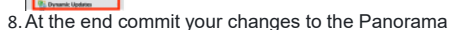


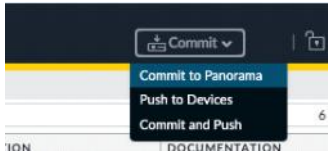
1. In the Panorama navigate to Device and select under Template your previous create Template (my example Student-TP)



6. As next click on the left panel on Dynamic Updates

7. Change the settings as shown in the picture below





#### 4. Deploy Firewalls in Azure

In the following chapter you will deploy several Software Firewalls in different PanOS version. The Software Firewalls will automatically join your previous created Panorama

##### 4.1 What you will do?

- Login to Azure Portal (<https://portal.azure.com>) and login with your Credentials
- Download Terraform Code from GitHub
- Modify Terraform Code
- Execute Terraform Code
- Validate Deployment in Azure Portal and Panorama

##### 4.2 Deployment

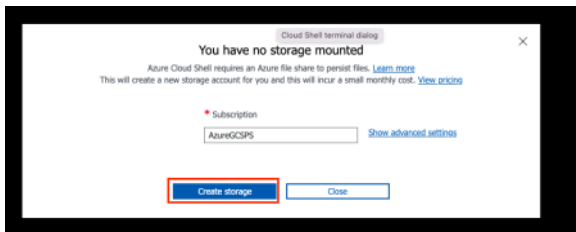
###### 1. Login in to Azure Portal (<https://portal.azure.com>)



###### 2. Open Azure Cloud Shell



3. click on Create storage. In some case it will not create a Storage Account. In that case click in "Show advanced settings" and create your own storage account.

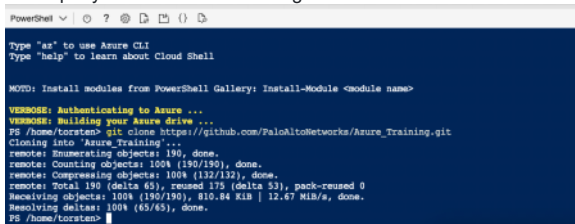


4. Once the creation of the storage is completed you will see the following



###### 5. Download Terraform Code from GitHub

1. in the Cloud shell execute the following command  
git clone <https://github.com/PaloAltoNetworks/flex-license-migration-lab.git>
2. As output you will see the following



6. Now browse to the deployment folder  
cd cd flex-license-migration-lab/azure/single\ firewall\ deployment/

7. Rename the terraform.tfvars.example to terraform.tfvars mv ./example.tfvars terraform.tfvars  
Expand For Details  
Command: mv ./terraform.tfvars.example terraform.tfvars

8. Modify the ``terraform.tfvars`` inside Cloud shell with the ``vi`` command 1. Modify the following variables in the File.  
resource\_group\_name = "migration-[Studentname]" #replace [Studentname] with your Name  
password = "SecurePassWord12!!" #change the password. Use a complex password  
storage\_account\_name = "panfstorage[Studentname]" #replace [Studentname] with your name in small letters without space  
storage\_share\_name = "bootstrapshare[Studentname]" #replace [Studentname] with your name in small letters without space

```
location = "north europe"
resource_group_name = "migration-test2"
common_vmseries_sku = "byol"
vm_series_version_set1 = "9.1.13"
vm_series_version_set2 = "18.8.9"
vm_series_version_set3 = "18.2.3"
username = "panadmin"
password = "SecurePassword121!"
allow_inbound_nsg_ip = [
 "0.0.0.0/0", # OR add your own public IP address here, visit "https://ifconfig.me/"
]
firewall1 = "Firewall1"
firewall2 = "Firewall2"
firewall3 = "Firewall3"
firewall4 = "Firewall4"
firewall5 = "Firewall5"
firewall6 = "Firewall6"
storage_account_name = "panstorage-test"
storage_share_name = "bootstrapshare-test"
azzones = ["1", "2", "3"]
files = {
 "files/authcodes" = "license/authcodes" # authcode is required only with common_vmseries_sku = "byol"
 "files/init-cfg.txt" = "config/init-cfg.txt"
}
```

1. Save your changes by pressing ESC and type :wq! and ENTER
2. As next switch to the folder files and rename the init-cfg.sample.txt to init-cfg.txt using the mv command
3. Modify the init-cfg.txt inside Cloud shell with the vi command. Make sure you added the same name of the Device Group and Template Stack you created in your Panorama. The value for the variables tplname and dname can be found in the section [3.3.2 License Panorama / Create Device Group and Template](#)

```
type=dhcp-client
vm-auth-key=123456789012345 #Follow the fLink below to create/show the key
panorama-server=10.1.2.3 #change it to the Public IP of your Panorama
tplname=my-stack #change it to the Template Stack inside your Panorama Section [3.3.2]
dname=my-device-group #change it to the Device Group inside your Panorama Section [3.3.2]
dhcp-send-hostname=yes
dhcp-send-client-id=yes
dhcp-accept-server-hostname=yes
dhcp-accept-server-domain=yes
```

[LINK to Guide for the Key creation](#)

1. As next in folder files and rename the authcodes.sample to authcodes using the mv command
2. Modify the authcodes files with the vi command.

XXXXXXX # Instructor will provide you the Key via Slack

1. Save your changes by pressing ESC and type :wq! and ENTER
2. Move back to the single\ firewall\ deployment folder with the command cd..
3. Once you made all your changes execute the Terraform code with following commands: 1. terraform init. 2. terraform plan. 3. terraform apply once you get the prompt type yes
4. Important! The complete deployment will take up to 10 Minutes after the completing the Terraform Apply. It is a good time for a break
5. Terraform Init

```
PS /home/torsten/Azure_Training/Azure_Autoscaling_Lab/vmseries_scaleset> terraform init
Initializing modules...
- inbound_bootstrap in ../modules/bootstrap
- inbound_lb in ../modules/loadbalancer
- inbound_scale_set in ../modules/vms
- outbound_lb in ../modules/loadbalancer
- vnet in ../modules/vnet

Initializing the backend...

Initializing provider plugins...
- Finding hashicorp/azurerm versions matching "~> 2.26, ~> 2.42, 2.64.0, ~> 2.64"...
- Finding hashicorp/random versions matching "~> 3.0"...
- Installing hashicorp/azurerm v2.64.0...
- Installed hashicorp/azurerm v2.64.0 (signed by HashiCorp)
- Installing hashicorp/random v3.4.3...
- Installed hashicorp/random v3.4.3 (signed by HashiCorp)

Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
PS /home/torsten/Azure_Training/Azure_Autoscaling_Lab/vmseries_scaleset>
```

## 6. Terraform Plan

```
Plan: 44 to add, 0 to change, 0 to destroy.

Changes to Outputs:
 + inbound_frontend_ips = {
 + frontend01 = (known after apply)
 }
 + metrics_instrumentation_key_inbound = (sensitive value)
 + password = (sensitive value)
 + username = "panadmin"
```

## 7. Terraform Apply

```
Do you want to perform these actions?
Terraform will perform the actions described above.
Only 'yes' will be accepted to approve.

Enter a value: yes
```

8. Once the terraform apply is completed you will see the following output

```
Apply complete! Resources: 44 added, 0 changed, 0 destroyed.

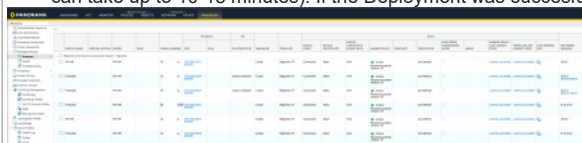
Outputs:
inbound_frontend_ips = {
 "frontend01" = "20.49"
}
metrics_instrumentation_key_inbound = <sensitive>
password = <sensitive>
username = "panadmin"
PS /home/torsten/Azure_Training/Azure_Autoscaling_Lab/vmseries_scaleset> |
```

#### 4.3 Validate Deployment

- Login into Panorama
- Validate Deployment

##### 1. Login into Panorama Public IP

2. Once you logged into the Panorama Navigate to the Panorama tab validate you can see your newly deployed Firewalls (The deployment and bootstrapping process can take up to 10-15 minutes). If the Deployment was successful you will see the following output in Panorama -> Managed Devices -> Summary



1. You successfully deployed your Environment if you can see the above output

Congratulations you successfully deployed several VM-Series Firewalls in different PanOS Version and bootstrapped them.

#### 5 License Migration

In the following steps you will migrate the previous created from a NON-Flex License model to the Flex-License model. You will do several migrations and create/update some Deployment profiles to fulfill the activities.

##### 5.1 Covered Scenarios in Detail

1. Migrate all Software Firewall to Flex License model (Fixed Deployment Profile) via Panorama
2. Migrate one (1) Firewall with PanOS 9.1.13-h3 to Fixed License via Panorama
3. Migrate one (1) Firewall with PanOS 10.0.9 to Fixed License via Panorama
4. Migrate one (1) Firewall with PanOS 10.0.9 from Fixed License to Flex via Panorama
5. Migrate one (1) Firewall with PanOS 10.2.3 to Fixed License via Panorama
6. Migrate one (1) Firewall with PanOS 10.2.3 from Fixed License to Flex via Panorama
7. Migrate one (1) Firewall with PanOS 10.2.3 from Flex to Flex with increasing the vCPU via Panorama

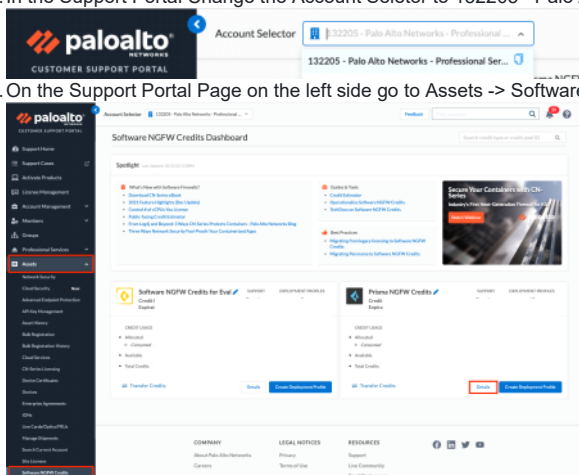
##### 5.2 Migrate Software Firewalls

In the following section we will migrate now all Software Firewall from the NON-Flex license model to the Flex license model. For that we will use the Deployment profile you created in the section [3.2 Create Fixed Deployment Profiles](#).

##### 5.2.1 Initial Migration

In the following section you migrate all Software Firewalls from NON-Flex Licensing to Flex Licensing.

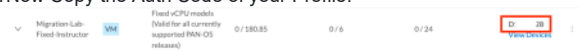
1. Login with your PANW Credentials at the Customer Support Portal <https://support.paloaltonetworks.com/>
2. In the Support Portal Change the Account Selector to 132205 - Palo Alto Networks - Professional Services



3. On the Support Portal Page on the left side go to Assets -> Software NGFW Credits -> Details

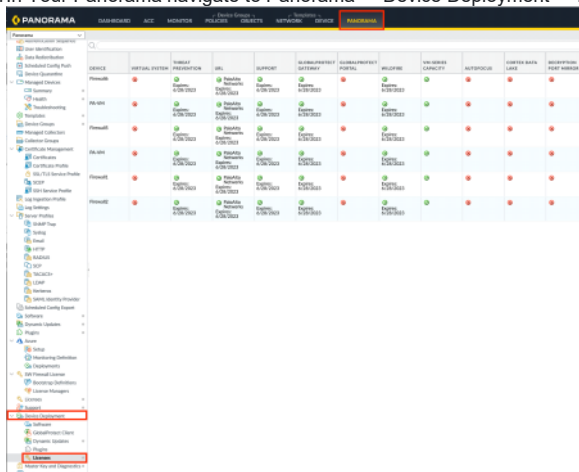
4. Now Search for your previous created Azure Deployment Profile [Here](#)

5. Now Copy the Auth Code of your Profile.



6. As next Login in to your Panorama [https://\[Public-IP\]](https://[Public-IP])

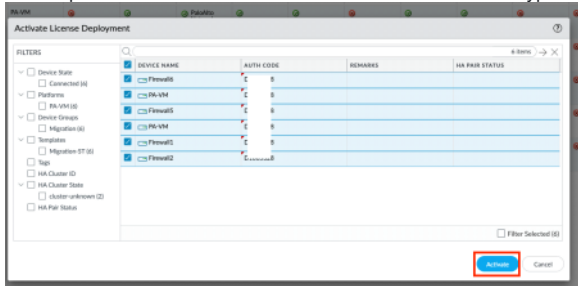
7. In Your Panorama navigate to Panorama -> Device Deployment -> Licenses



8. In the License window click at the bottom Activate

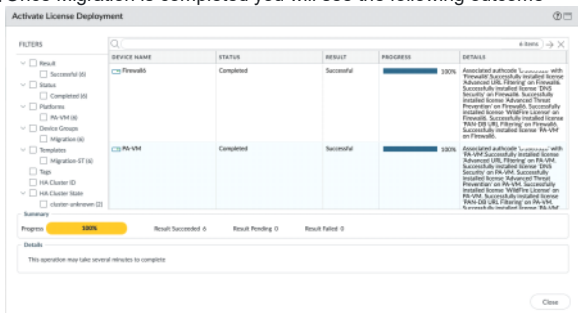


9. In the opened Window select now all available Firewalls and type in AUTH CODE field the auth code and click Activate



10. The Migration process will now take several minutes.

11. Once Migration is completed you will see the following outcome



12. As next check on the CSP if your credits got consumed from your deployment profile. You should see the below outcome

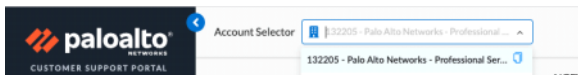


Congratulations!!! You Migrated successful all your Software Firewalls from a NON-Flex license model to Flex License model (Fixed Deployment Profile) via Panorama

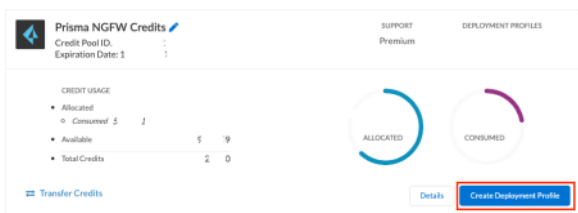
#### 5.2.2 Migrate PanOS 10.0.9 to Flexible Deployment Profile

In the following section we will create a new Deployment Profile to migrate the Software Firewalls from a Fixed Deployment Profile to a Flexible Deployment Profile

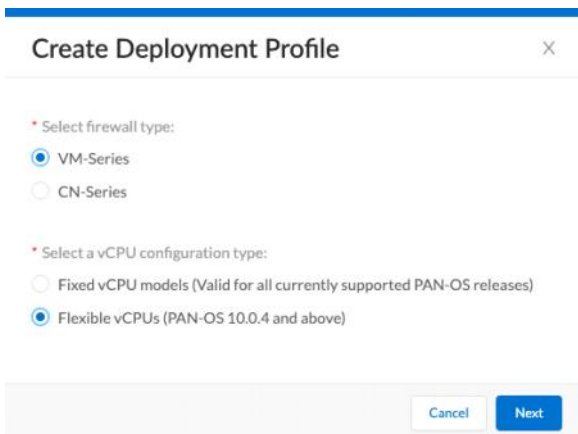
1. Login with your PANW Credentials at the Customer Support Portal <https://support.paloaltonetworks.com/>
2. In the Support Portal Change the Account Selector to 132205 - Palo Alto Networks - Professional Services



3. On the Support Portal Page on the left side go to Assets -> Software NGFW Credits
4. On the Prisma NFW Credits Pool click on Create Deployment Profile



5. Select the following as shown on the picture below and click Next



6. In the Deployment Profile use the following and use the NAME under "Profile Name" with "Migration-Lab-Flex-[StudentName]"

7. Click "Create Deployment Profile"

8. Verify that your Deployment Profile is successfully created

|                               |    |                                          |           |       |       |                              |
|-------------------------------|----|------------------------------------------|-----------|-------|-------|------------------------------|
| Migration Lab Flex Instructor | VM | Flexible vCPUs (VM: OS 50.0.4 and above) | 0 / 60.28 | 0 / 2 | 0 / 8 | <a href="#">View Devices</a> |
|-------------------------------|----|------------------------------------------|-----------|-------|-------|------------------------------|

9. Now Copy the Auth Code of the newly created Deployment Profile

10. As next Login in to your Panorama [https://\[Public-IP\]](https://[Public-IP])

11. In Your Panorama navigate to Panorama -> Device Deployment -> Licenses

12. In the License window click at the bottom Activate

13. Select now Firewall 3 and 4 or as in shown in the Picture the firewalls with the name "PA-VM".

You can verify the Name of the firewalls in the Summary tab. Now type in AUTH CODE field the auth code and click Activate

Expand For Details The Upgrade is not working because you changing the VM capacity. You can only upgrade the Software Firewalls to the Flex Deployment profile when activating it directly on the VM and using the "Upgrade VM capacity"

15. Before you can perform the License Key upgrade you have to install on the Software Firewalls the License API Key. Follow the [instructions](#) to perform the task. Repeat that

16. Install API License Key on ALL other Software Firewalls (1-6) too for future tasks

17. Once you added the API go in your Panorama and switch the context to Firewall 3 or 4 (or PA-VM)

18. In the Firewall navigate Device -> License and click on Upgrade VM capacity

19. In the window add under Authorization Code your atuh code and click Continue



**Upgrade VM Capacity** ⓘ

You are upgrading VM-X to VM-Y with subscriptions listed below

**SUBSCRIPTIONS**

GlobalProtect Gateway  
PAN-DB URL Filtering  
DNS Security  
WildFire License  
PA-VM  
Advanced URL Filtering

**Upgrade Options**

☒ Authorization Code ☐ License Key ☐ Fetch from license server

Authorization Code **D 13**

Upon upgrading the VM-Y will retain all the configurations of VM-X and would have higher capacity. Identical subscriptions should be applied for VM-Y for maintaining firewall functionality available from those subscriptions.

Click Continue and PanOS will upgrade to VM-Y.

Click Complete Manually if PanOS does not have access to the license server. A license upgrade file will be created and you will be prompted to save it to a machine that can access the license server.

20. You will see the below outcome once it completed. Click close and refresh the UI

**Upgrade VM Capacity** ⓘ

Upgraded license to VM-FLEX-4 and deactivated old licenses.

Please refresh the UI after the services are up.

21. In the Firewall switch to the Dashboard and you can see the VM License changed to VM-FLEX-4

**General Information** ⓘ

Device Name: PA-VM  
MGT IP Address: 192.168.100.7 (DHCP)  
MGT Netmask: 255.255.255.128  
MGT Default Gateway: 192.168.100.1  
MGT IPv6 Address: unknown  
MGT IPv6 Link Local Address: fe80::20d:3aff:feba:8398/64  
MGT IPv6 Default Gateway: 0000:3a:ba:83:98  
MGT MAC Address: 00:00:3a:ba:83:98  
Model: PA-VM  
Serial #: 0 26  
CPU ID: A FFB0B1F  
UUID: 9: ~3D48-9C0D-A1AABEE08551  
VM Cores: 4  
VM Memory: 14363676  
VM License: **VM-FLEX-4**  
VM Capacity: 9.0 GB  
VM Mode: Microsoft Azure  
Software Version: 10.0.9  
GlobalProtect Agent: 0.0.0  
Application Version: 8518-7198  
Threat Version: 8518-7198  
Device Dictionary Version: 76-393 (05/05/23)  
URL Filtering Version: 20230510.20170  
GlobalProtect Clientless VPN Version: 0  
Time: Wed May 10 04:18:42 2023  
Uptime: 0 days, 5:01:45  
Plugin DLP: dlp-1.0.3  
Plugin VM-Series: vm\_series-2.1.4  
Device Certificate Status: None

22. Repeat the same steps for the second firewall.

23. When you know go to the Support Portal and check your profiles, you can see that the count of the Fixed prile is reduced by 2 firewalls and 8 vcpus and the Flex profile increased.

|                                  |                                                                      |                 |       |         |                              |
|----------------------------------|----------------------------------------------------------------------|-----------------|-------|---------|------------------------------|
| Migration Lab - Fixed Instructor | Fixed-CPU models (Valid for all currently supported PAN-OS releases) | 130/56 / 190/85 | 4 / 6 | 16 / 24 | <a href="#">View Devices</a> |
| Migration Lab - Flex Instructor  | Flexible-CPU (PAN-OS 10.0.4 and above)                               | 60/28 / 60/28   | 2 / 2 | 8 / 8   | <a href="#">View Devices</a> |

Congratulations!!! You successful migrated 2 Firewalls from a Fixed License Deployment Profile to an Flexible Deployment profile and implemented the API License Key on the Firewalls

### 5.2.3 Migrate PanOS 10.2.3 to Flexible Deployment Profile

In the following section we will create a new Deployment Profile to migrate the Software Firewalls from a Fixed Deployment Profile to a Flexible Deployment Profile

1. Login with your PANW Credentials at the Customer Support Portal <https://support.paloaltonetworks.com/>
2. In the Support Portal Change the Account Selector to 132205 - Palo Alto Networks - Professional Services

**paloalto** CUSTOMER SUPPORT PORTAL

Account Selector **132205 - Palo Alto Networks - Professional Services**

3. On the Support Portal Page on the left side go to Assets -> Software NGFW Credits
4. On the Prisma VM NFGW Credits Pool click on Create Deployment Profile



Prisma NGFW Credits

Credit Pool ID: ...

Expiration Date: 1 ...

SUPPORT: Premium

DEPLOYMENT PROFILES

CREDIT USAGE

- Allocated: 5
- Consumed: 1
- Available: 4
- Total Credits: 20

ALLOCATED CONSUMED

Transfer Credits

Details Create Deployment Profile

5. Select the following as shown on the picture below and click Next

### Create Deployment Profile

Select firewall type:

☒ VM-Series

☐ CN-Series

Select a vCPU configuration type:

☐ Fixed vCPU models (Valid for all currently supported PAN-OS releases)

☒ Flexible vCPUs (PAN-OS 10.0.4 and above)

Cancel Next

6. In the Deployment Profile use the following and use the NAME under "Profile Name" with "Migration-Lab-Flex-10.2-[StudentName]"

### Create Deployment Profile

VM-Series

Profile Name: Migration-Lab-Flex-10.2-Instructor

Number of Firewalls: 2

Planned vCPU per Firewall: 4

Security Use Case: Custom

Customize Subscriptions

☒ Advanced URL Filtering

☐ Intelligent Traffic Offload

☒ DNS

☒ Advanced Threat Prevention

☐ Global Protect

☐ Web Proxy (Promotional Offer)

☐ DLP

☒ Advanced Wildfire

☐ SD-WAN

Legacy Subscriptions

Use Credits to Enable VM Panorama

☒ For Management

☐ As Dedicated Log Collector

Protect more, save more

Calculate Estimated Cost

Cancel Create Deployment Profile

7. Click "Create Deployment Profile"

8. Verify that your Deployment Profile is successfully created

| PROFILE NAME                       | FIREWALL TYPE | PAN-OS VERSION                           | CREDITS CONSUMED/ALLOCATED | FIREWALLS DEPLOYED/PLANNED | VCPU CONSUMED/ALLOCATED | AUTH CODE |
|------------------------------------|---------------|------------------------------------------|----------------------------|----------------------------|-------------------------|-----------|
| Migration-Lab-Flex-10.2-Instructor | VM            | Flexible vCPUs (PAN-OS 10.0.4 and above) | 0/0 Credits Pending        | 0/2                        | 0/8                     | ---       |

9. Now Copy the Auth Code of the newly created Deployment Profile

10. As next Login in to your Panorama [https://\[Public-IP\]](https://[Public-IP])

11. In Your Panorama navigate to Panorama -> Device Deployment -> Licenses

PANORAMA

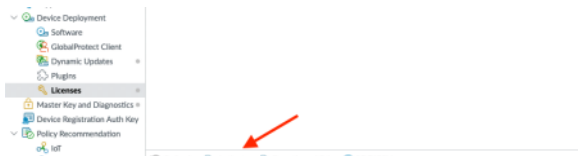
Device Deployment

Licenses

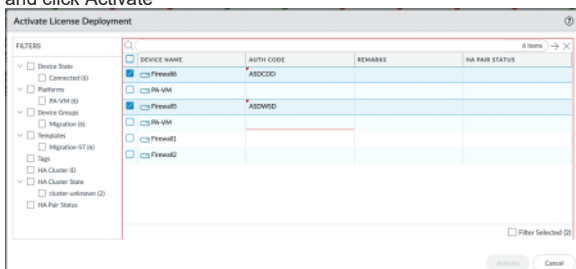
| Name | Status | Version | ... |
|------|--------|---------|-----|
| ...  | ...    | ...     | ... |

Activate

12. In the License window click at the bottom Activate



13. Select now Firewall 5 and 6 or as in shown in the Picture. You can verify the Name of the firewalls in the Summary tab. Now type in AUTH CODE field the auth code and click Activate

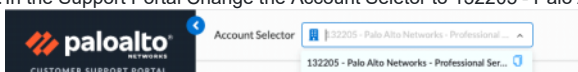


14. It will fail too because of the same issue you already faced above. Please follow the same instructions from the previous chapter to migrate the firewalls to Flexible Deployment profile.

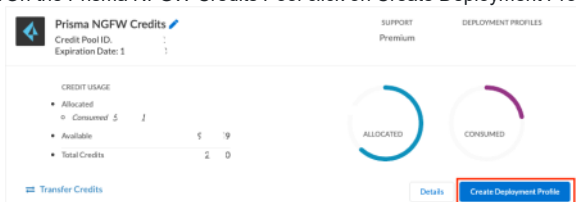
### 5.3 Change vCPU on PanOS 10.2.3 Firewall

In the following section we will create a new Deployment Profile to change the vCPU on the already licensed Software Firewall

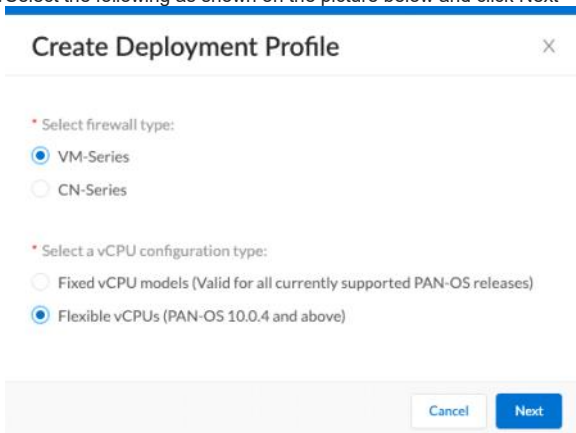
1. Login with your PANW Credentials at the Customer Support Portal <https://support.paloaltonetworks.com/>
2. In the Support Portal Change the Account Selector to 132205 - Palo Alto Networks - Professional Services



3. On the Support Portal Page on the left side go to Assets -> Software NGFW Credits
4. On the Prisma NGFW Credits Pool click on Create Deployment Profile



5. Select the following as shown on the picture below and click Next



6. In the Deployment Profile use the following and use the NAME under "Profile Name" with "Migration-Lab-Flex-10.2-3vcpu-{StudentName}"

## Create Deployment Profile ✕

**VM-Series**

Profile Name

\* Number of Firewalls

\* Planned vCPU per Firewall

\* Security Use Case

Customize Subscriptions

☒ Advanced URL Filtering ☐ Intelligent Traffic Offload

☒ DNS ☒ Advanced Threat Prevention

☐ Global Protect ☐ Web Proxy (Promotional Offer)

☐ DLP ☒ Advanced Wildfire

☐ SD-WAN

> Legacy Subscriptions

Use Credits to Enable VM Panorama

☒ For Management ☐ As Dedicated Log Collector

Protect more, save more

[Calculate Estimated Cost](#)

[Cancel](#) [Create Deployment Profile](#)

1. Click "Create Deployment Profile"
2. Verify that your Deployment Profile is successfully created

|                                          |    |                                          |       |       |       |                              |
|------------------------------------------|----|------------------------------------------|-------|-------|-------|------------------------------|
| Migration-Lab-Flex-10.2-3vcpu-Instructor | VM | Flexible vCPUs (PAN-OS 10.0.4 and above) | 0 / 0 | 0 / 1 | 0 / 3 | <a href="#">View Devices</a> |
|------------------------------------------|----|------------------------------------------|-------|-------|-------|------------------------------|

1. Verify at first that both software Firewalls (5 and 6) are migrated to the new Flexible Deployment Profile. Check the Firewall Dashboard if you can see (VM-Series-4)

### General Information

|                             |                             |
|-----------------------------|-----------------------------|
| Device Name                 | Firewall6                   |
| MGT IP Address              | 192.168.100.4 (DHCP)        |
| MGT Netmask                 | 255.255.255.128             |
| MGT Default Gateway         | 192.168.100.1               |
| MGT IPv6 Address            | unknown                     |
| MGT IPv6 Link Local Address | fe80::20d:3aff:feba:8ad9/64 |
| MGT IPv6 Default Gateway    |                             |
| MGT MAC Address             | 00:0d:3a:ba:8a:d9           |
| Model                       | PA-VM                       |
| Serial #                    | 0C-----25                   |
| CPU ID                      | A2 FB8B1F                   |
| UUID                        | 84 :DC4E-AFBA-28CEE1B774E0  |
| VM Cores                    | 4                           |
| VM Memory                   | 14351728                    |
| VM License                  | VM-SERIES-4                 |
| VM Capacity Tier            | 9.0 GB                      |
| VM Mode                     | Microsoft Azure             |
| Software Version            | 10.2.3                      |
| GlobalProtect Agent         | 0.0.0                       |
| Application Version         | 8710-8054 (05/15/23)        |
| Threat Version              | 8710-8054 (05/15/23)        |
| Antivirus Version           | 4451-4968 (05/15/23)        |
| Device Dictionary Version   | 77-395 (05/11/23)           |
| WildFire Version            | 768754-772228 (05/15/23)    |
| URL Filtering Version       | 20230516.20088              |

1. As next login to Firewall 5 or 6 via ssh. In my Example i migrate Firewall 6  
ssh -oHostKeyAlgorithms=+ssh-rsa USERNAME@FIREWALL IP
2. In the CLI type the following command to set the Core value to 3  
request plugins vm\_series set-cores cores 3

```
panadmin@Firewall6> request plugins vm_series set-cores cores 3

Total cores configured to 3.
Setting custom cores requires system reboot. Custom DP cores might get reset to accomodate the new custom core setting.

panadmin@Firewall6>
```

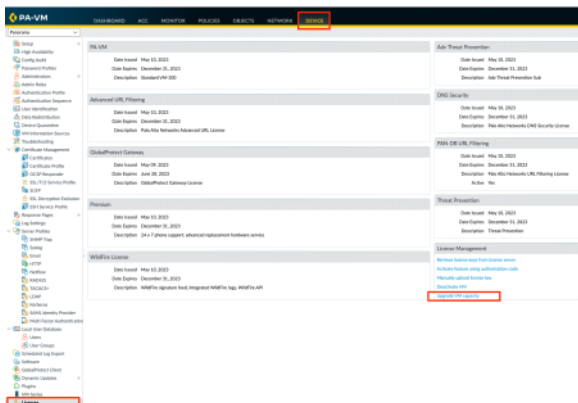
1. The requires a reboot. Type the following command to rebboot the Firewall request restart system

```
panadmin@Firewall16> request restart system
Executing this command will disconnect the current session. Do you want to continue? (y or n)

Broadcast message from root (pts/1) (Mon May 15 23:01:05 2023):

The system is going down for reboot NOW!
```

2. The Reboot of the firewall will take now around ~ 5 Minutes
3. Once the Firewall is back online and function login to the Firewall via Panorama or directly to the Firewall
4. In the Firewall navigate Device -> License and click on Upgrade VM capacity



5. In the window add under Authorization Code your auth code (3 vCPU) and click Continue

### Upgrade VM Capacity

You are upgrading VM-X to VM-Y with subscriptions listed below

**SUBSCRIPTIONS**

- GlobalProtect Gateway
- PAN-DB URL Filtering
- DNS Security
- WildFire License
- PA-VM
- Advanced URL Filtering

**Upgrade Options**

☒ Authorization Code
 ☐ License Key
 ☐ Fetch from license server

Authorization Code:

Upon upgrading the VM-Y will retain all the configurations of VM-X and would have higher capacity. Identical subscriptions should be applied for VM-Y for maintaining firewall functionality available from those subscriptions.

Click Continue and PanOS will upgrade to VM-Y.

Click Complete Manually if PanOS does not have access to the license server. A license upgrade file will be created and you will be prompted to save it to a machine that can access the license server.

6. You will see the below outcome once it completed. Click close and refresh the UI

### Upgrade VM Capacity

Upgraded license to VM-FLEX-3 and deactivated old licenses.

Please refresh the UI after the services are up.

7. In the Firewall go to the Dashboard and you can see the VM License changed to VM-FLEX-3

| General Information         |                             |
|-----------------------------|-----------------------------|
| Device Name                 | Firewall6                   |
| MGT IP Address              | 192.168.100.4 (DHCP)        |
| MGT Netmask                 | 255.255.255.128             |
| MGT Default Gateway         | 192.168.100.1               |
| MGT IPv6 Address            | unknown                     |
| MGT IPv6 Link Local Address | fe80::20d:3aff:feba:8ad9/64 |
| MGT IPv6 Default Gateway    |                             |
| MGT MAC Address             | 00:0d:3a:ba:8ad9            |
| Model                       | PA-VM                       |
| Serial #                    | 0x                          |
| CPU ID                      | A:                          |
| UUID                        | 84 4E0                      |
| VM Cores                    | 3                           |
| VM Memory                   | 14351728                    |
| VM License                  | VM-SERIES-3                 |
| VM Capacity Tier            | T2-14GB                     |
| VM Mode                     | Microsoft Azure             |
| Software Version            | 10.2.3                      |
| GlobalProtect Agent         | 0.0.0                       |
| Application Version         | 8710-8054 (05/15/23)        |
| Threat Version              | 8710-8054 (05/15/23)        |
| Antivirus Version           | 4451-4968 (05/15/23)        |
| Device Dictionary Version   | 77-395 (05/11/23)           |
| WildFire Version            | 768766-772240 (05/15/23)    |
| URL Filtering Version       | 20230516.20114              |

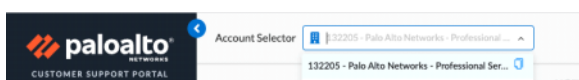
Congratulations!!! You successful migrated 1 Software Firewalls from Flexible Deployment Profile with 4 vCPU's to a Flexible Deployment Profile and changed the Cores count via CLI

#### 5.4 Change/Update Deployment Profiles

In the following section you will now update your Deployment Profile ("Migration-Lab-Flex-10.2-[StudentName]") too remove some Subscription and enable subscriptions

##### 5.4.1 Add Security Subscriptions

1. Login with your PANW Credentials at the Customer Support Portal <https://support.paloaltonetworks.com/>
2. In the Support Portal Change the Account Seletor to 132205 - Palo Alto Networks - Professional Services



3. On the Support Portal Page on the left side go to Assets -> Software NGFW Credits
4. Go to youe Deployment Profile "Migration-Lab-Flex-10.2-[StudentName]" click in the three dots and Edit Profile

Migration-Lab-Flex-10.2-Instructor

VM

Flexible vCPUs (PAN-OS 10.0.4 and above)

32.59 / 65.18

1 / 2

4 / 8

D6183727

View Devices

COMPANY

About Palo Alto Networks

Careers

LEGAL NOTICES

Privacy

Terms of Use

RESOURCES

Support

Live Community

Email Preferences

Learning Center (Beacon)

Technical Documentation

f

in

tw

yt

Register Firewall

Provision Panorama

Edit Profile

Delete

Transfer Profile

Deactivate Firewall

Clone Profile

5. In your Deployment Profile select the Global Protect and Click Update Deployment Profile

### Edit Deployment Profile

VM-Series

Profile Name: Migration-Lab-Flex-10.2-Instructor

Number of Firewalls: 2

Planned vCPU per Firewall: 4

Security Use Case: Custom

Customize Subscriptions

- ☒ Advanced URL Filtering
- ☒ DNS
- ☒ Global Protect
- ☐ DLP
- ☐ SD-WAN
- ☐ Intelligent Traffic Offload
- ☒ Advanced Threat Prevention
- ☐ Web Proxy (Promotional Offer)
- ☒ Advanced Wildfire

Legacy Subscriptions

Use Credits to Enable VM Panorama

- ☒ For Management
- ☐ As Dedicated Log Collector

Protect more, save more

[Calculate Estimated Cost](#)

[Cancel](#) [Update Deployment Profile](#)

6. Click YES in the new Window

### Edit Deployment Profile

Any change in security subscriptions will affect all the deployed firewalls. Are you sure?

[No](#) [Yes](#)

7. As next Login to your Panorama

8. In Your Panorama navigate to Panorama -> Device Deployment -> Licenses

PANORAMA

Home | Configuration | Reports | Tools | Settings | Help

Configuration

- Device Deployment
  - Licenses

| Device | Serial               | Model   | Version | License              | License Type | License Status | License Key          | License Expiry |
|--------|----------------------|---------|---------|----------------------|--------------|----------------|----------------------|----------------|
| FW-001 | 10000000000000000000 | PA-7000 | 7.0.0   | 10000000000000000000 | VM-Series    | Active         | 10000000000000000000 | 2023-12-31     |
| FW-002 | 10000000000000000000 | PA-7000 | 7.0.0   | 10000000000000000000 | VM-Series    | Active         | 10000000000000000000 | 2023-12-31     |
| FW-003 | 10000000000000000000 | PA-7000 | 7.0.0   | 10000000000000000000 | VM-Series    | Active         | 10000000000000000000 | 2023-12-31     |
| FW-004 | 10000000000000000000 | PA-7000 | 7.0.0   | 10000000000000000000 | VM-Series    | Active         | 10000000000000000000 | 2023-12-31     |
| FW-005 | 10000000000000000000 | PA-7000 | 7.0.0   | 10000000000000000000 | VM-Series    | Active         | 10000000000000000000 | 2023-12-31     |

9. Select Refresh

[Refresh](#) [Activate](#) [Deactivate VMs](#) [PDF/CSV](#)

10. Now Select the firewall who was associated with the Auth Code of the "Migration-Lab-Flex-10.2-[StudentName]" Deployment Profile (In the Example is it Firewall 5) and click Refresh

### Refresh License Deployment

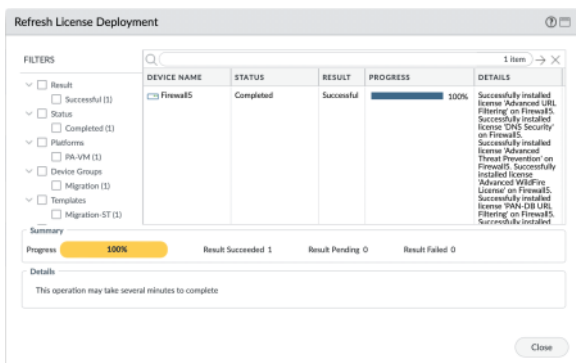
Filters

- ☒ Device State
  - ☐ Connected (0)
- ☒ Platforms
  - ☐ PA-VM (0)
- ☒ Device Groups
  - ☐ Migration (0)
- ☒ Tags
  - ☐ Migration-07 (0)
- ☒ HA Cluster ID
  - ☐ Cluster unknown (0)
- ☒ HA Pair Status
  - ☐ HA Pair Status (0)

| Device Name | Remarks | HA Pair Status |
|-------------|---------|----------------|
| FW-001      |         |                |
| FW-002      |         |                |
| FW-003      |         |                |
| FW-004      |         |                |
| FW-005      |         |                |
| FW-006      |         |                |

[Refresh](#) [Cancel](#)

11. You should see the following output if it was successful



12. Refresh the Panorama UI

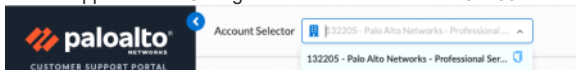
13. Now you should see that on Software Firewall 5 is the Global Protect License Active

| DEVICE    | VERSION | THREAT PREVENTION | URL     | SUPPORT | GLOBAL PROTECT | GLOBAL PROTECT | WILDFIRE | VM SERIES | AUTOFOCUS | CORRELATION | DESCRIPTION                   |
|-----------|---------|-------------------|---------|---------|----------------|----------------|----------|-----------|-----------|-------------|-------------------------------|
| Firewall5 | 10.2.3  | Enabled           | Enabled | Enabled | Enabled        | Enabled        | Enabled  | Enabled   | Enabled   | Enabled     | Global Protect License Active |
| FW-104    | 10.2.3  | Enabled           | Enabled | Enabled | Enabled        | Enabled        | Enabled  | Enabled   | Enabled   | Enabled     | Global Protect License Active |
| Firewall5 | 10.2.3  | Enabled           | Enabled | Enabled | Enabled        | Enabled        | Enabled  | Enabled   | Enabled   | Enabled     | Global Protect License Active |
| FW-104    | 10.2.3  | Enabled           | Enabled | Enabled | Enabled        | Enabled        | Enabled  | Enabled   | Enabled   | Enabled     | Global Protect License Active |
| Firewall5 | 10.2.3  | Enabled           | Enabled | Enabled | Enabled        | Enabled        | Enabled  | Enabled   | Enabled   | Enabled     | Global Protect License Active |
| FW-104    | 10.2.3  | Enabled           | Enabled | Enabled | Enabled        | Enabled        | Enabled  | Enabled   | Enabled   | Enabled     | Global Protect License Active |

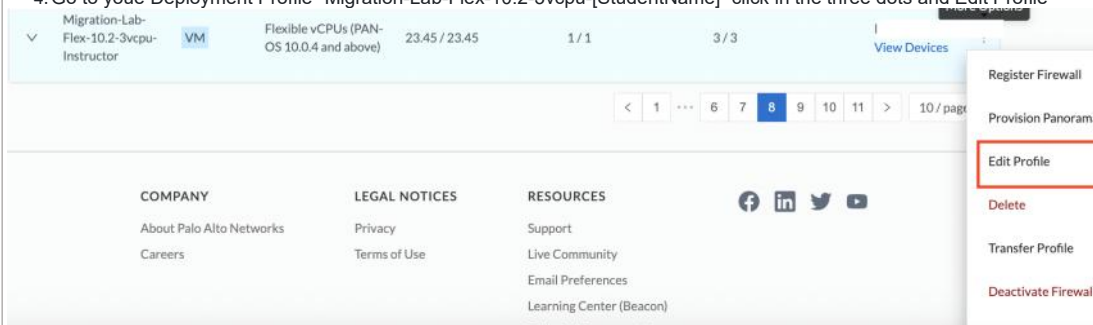
Congratulations!!! You successful Updated your Deployment Profile and added another Security subscription

#### 5.4.2 Remove Security Subscriptions

1. Login with your PANW Credentials at the Customer Support Portal <https://support.paloaltonetworks.com/>
2. In the Support Portal Change the Account Selector to 132205 - Palo Alto Networks - Professional Services



3. On the Support Portal Page on the left side go to Assets -> Software NGFW Credits
4. Go to your Deployment Profile "Migration-Lab-Flex-10.2-3vcpu-[StudentName]" click in the three dots and Edit Profile



1. In your Deployment Profile de-select the DNSSt and Click Update Deployment Profile

1. Click YES in the new Window

## 2. As next Login to your Panorama

3. In Your Panorama navigate to Panorama -> Device Deployment -> Licenses

#### 4. Select Refresh

5. Now Select the Firewall who is associated with the Update Auth Code.



Refresh License Deployment

FILTERS

Device State

☐ Connected (6)

Platforms

☐ PA-VM (6)

Device Groups

☐ Migration (6)

Templates

☐ Migration-ST (6)

Tags

☐ HA Cluster ID

HA Cluster State

☐ cluster-unknown (2)

HA Pair Status

☐

6 items

→

×

| <input type="checkbox"/>            | DEVICE NAME | REMARKS | HA PAIR STATUS |
|-------------------------------------|-------------|---------|----------------|
| <input checked="" type="checkbox"/> | Firewall6   |         |                |
| <input type="checkbox"/>            | PA-VM       |         |                |
| <input type="checkbox"/>            | Firewall5   |         |                |
| <input type="checkbox"/>            | PA-VM       |         |                |
| <input type="checkbox"/>            | Firewall1   |         |                |
| <input type="checkbox"/>            | Firewall2   |         |                |

☐ Filter Selected (1)

Refresh

Cancel

1. Are the Update working?

2. If everything Worked you have to refresh the UI. In some cases it will show you an error but when you check the Panorama you can see the DNS license got removed or is listed as expired

| DEVICE    | VIRTUAL SYSTEM | THREAT PREVENTION   | URL                                   | SUPPORT             | GLOBALPROT... GATEWAY | GLOBALPROT... PORTAL | WILDFIRE            | VM-SERIES CAPACITY  | AUTOFOCUS | CORTEX DATA LAKE | DECRYPTION PORT MIRROR | NETWORK PACKET BROKER | DNS SECURITY                       |
|-----------|----------------|---------------------|---------------------------------------|---------------------|-----------------------|----------------------|---------------------|---------------------|-----------|------------------|------------------------|-----------------------|------------------------------------|
| Firewall6 |                | Expires: 12/31/2023 | PaloAlto Networks Expires: 12/31/2023 | Expires: 12/31/2023 |                       |                      | Expires: 12/31/2023 | Expires: 12/31/2023 |           |                  |                        |                       | License Expired Expires: 5/16/2023 |

Congratulations!!! You successful Updated your Deployment Profile and removed a Security subscription

From <<https://github.com/PaloAltoNetworks/flex-license-migration-lab>>

Palo-Alto Firewall Page 25

# Palo-Alto VM using VMWare Workstation

Thursday, December 25, 2025 8:40 PM

OVA Download link: <https://upw.io/4A9/PA-VM-ESX-10.1.0.ova>

## Hypervisor

- VMWare workstation (latest version)
- Oracle Virtual Box (latest version)

Default username: **admin**

Default password: **admin** (note - you might have to login 4-5 times with the default password)

To view the system details

# show system info

Other commands:

### Basic System Commands

```
show system info
show system state
show system environmentals
show system resources
show system disk-space
show system software status
show system uptime
show system files
```

### Interface & Networking Commands

```
show interface all
show interface ethernet1/1
show interface management
show routing route
show routing protocol bgp summary
show routing protocol ospf neighbor
show arp all
show arp interface ethernet1/1
ping host <ip-address>
traceroute host <ip-address>
```

### Policy & Security Rule Commands

```
show running security-policy
show running nat-policy
show running qos-policy
show running decryption-policy
show running application-override-policy
```

### Traffic, Session & Application Commands

show session all  
 show session id <session-id>  
 show session info  
 show session statistics  
 show counter global  
 show counter interface  
 show application all  
 show application statistics

Accessing PA using web browser:

General Information

|                                      |                                     |
|--------------------------------------|-------------------------------------|
| Device Name                          | PA-VM                               |
| MGT IP Address                       | 192.168.1.9 (DHCP)                  |
| MGT Netmask                          | 255.255.255.0                       |
| MGT Default Gateway                  | 192.168.1.1                         |
| MGT IPv6 Address                     | unknown                             |
| MGT IPv6 Link Local Address          | fe80::20c:29ff:fe14:1147/64         |
| MGT IPv6 Default Gateway             |                                     |
| MGT MAC Address                      | 00:0c:29:14:11:47                   |
| Model                                | PA-VM                               |
| Serial #                             | unknown                             |
| CPU ID                               | ESX:52060A00FFB8B1F                 |
| UUID                                 | 564D080B-CFFA-26BA-D1F2-D6F74141147 |
| VM Cores                             | 2                                   |
| VM Memory                            | 5580648                             |
| VM License                           | none                                |
| VM Capacity Tier                     | unknown                             |
| VM Mode                              | VMware ESXi                         |
| Software Version                     | 10.1.0                              |
| GlobalProtect Agent                  | 0.0.0                               |
| Application Version                  | 8402-6681                           |
| Threat Version                       | 8402-6681                           |
| Device Dictionary Version            | 1-211                               |
| URL Filtering Version                | 0000.00.00.000                      |
| GlobalProtect Clientless VPN Version | 0                                   |
| Time                                 | Thu Dec 25 13:28:06 2025            |

Logged In Admins

| Admin | From    | Client | Session Start  | Idle For  |
|-------|---------|--------|----------------|-----------|
| admin | Console | CU     | 12/25 13:21:42 | 00:06:06s |

Welcome

## Welcome to PAN-OS 10.1!

With this release, Palo Alto Networks introduces an integrated CASB (Cloud Access Security Broker) solution to enable SaaS applications with confidence, and a reinvention of Internet security with the introduction of Advanced URL Filtering and major enhancements to our DNS Security service. Highlights include:

**App-ID Cloud Engine**—Powers our SaaS Security Inline subscription and dramatically increases visibility and control of over 15,000 SaaS applications and their corresponding functions. Applications identified through the App-ID Cloud Engine integrate with Policy Optimizer to streamline incorporation of these new applications into an application-based security policy with the strongest possible security posture. New applications become available in PAN-OS through the App-ID Cloud Engine as they're defined, without needing to wait for App-ID signatures to be developed.

**SaaS Security Inline Subscription**—In combination with next-generation firewalls, the SaaS Security Inline subscription provides SaaS visibility and security controls that prevent data security risks of unsanctioned SaaS app traffic traversing your network. With SaaS policy rule recommendations authored by the SaaS administrator and imported as policy rules by the firewall administration, SaaS Security Inline facilitates a seamless SaaS security workflow throughout your organization for an improved security posture.

**Advanced URL Filtering**—Extends beyond basic URL filtering with the introduction of a new, cloud-based ML-powered web security engine that protects against today's most evasive and targeted web-based attacks. Advanced URL Filtering performs ML-based inspection of real web traffic inline to discover malicious behavior as it flows through the network and out of-band web resources to detect and prevent advanced file-based web-based attacks including malware, ransomware, and data exfiltration.

☒ Do not show again

Close

Dashboard:

General Information

|                                      |                                     |
|--------------------------------------|-------------------------------------|
| Device Name                          | PA-VM                               |
| MGT IP Address                       | 192.168.1.9 (DHCP)                  |
| MGT Netmask                          | 255.255.255.0                       |
| MGT Default Gateway                  | 192.168.1.1                         |
| MGT IPv6 Address                     | unknown                             |
| MGT IPv6 Link Local Address          | fe80::20c:29ff:fe14:1147/64         |
| MGT IPv6 Default Gateway             |                                     |
| MGT MAC Address                      | 00:0c:29:14:11:47                   |
| Model                                | PA-VM                               |
| Serial #                             | unknown                             |
| CPU ID                               | ESX:52060A00FFB8B1F                 |
| UUID                                 | 564D080B-CFFA-26BA-D1F2-D6F74141147 |
| VM Cores                             | 2                                   |
| VM Memory                            | 5580648                             |
| VM License                           | none                                |
| VM Capacity Tier                     | unknown                             |
| VM Mode                              | VMware ESXi                         |
| Software Version                     | 10.1.0                              |
| GlobalProtect Agent                  | 0.0.0                               |
| Application Version                  | 8402-6681                           |
| Threat Version                       | 8402-6681                           |
| Device Dictionary Version            | 1-211                               |
| URL Filtering Version                | 0000.00.00.000                      |
| GlobalProtect Clientless VPN Version | 0                                   |
| Time                                 | Thu Dec 25 13:32:36 2025            |
| Uptime                               | 0 days, 0:16:22                     |
| Plugin VM-Series                     | vm-series-2.1.0                     |
| Plugin DLP                           | dlp-1.0.3                           |
| Device Certificate Status            | None                                |

Logged In Admins

| Admin | From        | Client | Session Start  | Idle For  |
|-------|-------------|--------|----------------|-----------|
| admin | Console     | CU     | 12/25 13:21:42 | 00:10:36s |
| admin | Console     | CU     | 12/25 13:22:11 | 00:10:15s |
| admin | Console     | CU     | 12/25 13:22:33 | 00:02:18s |
| admin | 192.168.1.8 | Web    | 12/25 13:27:56 | 00:00:00s |

Config Logs

No data available.

Locks

No locks found

ACC Risk Factor (Last 60 minutes)

0.0

System Logs

| Description                                                | Time           |
|------------------------------------------------------------|----------------|
| MLAV: Unknown error.                                       | 12/25 13:30:41 |
| User admin logged in via Web from 192.168.1.8 using https  | 12/25 13:27:56 |
| authenticated for user 'admin'. From: 192.168.1.8.         | 12/25 13:27:56 |
| MLAV: Unknown error.                                       | 12/25 13:26:10 |
| LOGIN ON try1 BY admin                                     | 12/25 13:22:56 |
| FAILED LOGIN 1 FROM try1 FOR admin. Authentication failure | 12/25 13:22:56 |
| Password changed for user admin.                           | 12/25 13:22:43 |
| authenticated for user 'admin'.                            | 12/25 13:22:43 |
| User admin logged in via CLI from Console                  | 12/25 13:22:32 |
| authenticated for user 'admin'. From: Console or telnet.   | 12/25 13:22:32 |

# Palo-Alto VM using Oracle VirtualBox

Sunday, December 28, 2025

8:57 AM

## 1. Prerequisites and Planning

### 1.1 Supported Palo Alto VM Image

Use VM-Series for VMware (OVA)

(Palo Alto does not officially provide a VirtualBox image; VirtualBox works but is best-effort only).

### 1.2 Host System Requirements

- CPU with hardware virtualization enabled (Intel VT-x / AMD-V)
- Minimum 8 GB RAM (16 GB recommended)
- At least 60 GB free disk space
- Oracle VirtualBox 7.x (latest recommended)
- VirtualBox Extension Pack installed

### 1.3 Files Required

Palo Alto VM-Series .ova file

Internet access (optional but recommended for updates)

## 2. Importing the Palo Alto OVA into VirtualBox

### Step 1: Launch VirtualBox

- Open **Oracle VirtualBox Manager**

### Step 2: Import Appliance

- a. Click **File → Import Appliance**
- b. Browse and select the **Palo Alto .ova file**
- c. Click **Next**

### Step 3: Review and Modify Appliance Settings

Carefully validate and adjust the following:

### 3.1 System → Motherboard

- **Base Memory:** 8192 MB (minimum)
- **Boot Order:** Hard Disk first
- **Chipset:** PIIX3
- **EFI:** **X** Disabled

### 3.2 System → Processor

- **CPUs:** 4 (minimum)
- **Enable PAE/NX:** ☒ Enabled

### 3.3 Display

- **Graphics Controller:** VMSVGA
- **Video Memory:** 128 MB

Avoid VBoxVGA or VBoxSVGA — they often cause boot instability.

### 3.4 Storage

- Confirm the **virtual disk is attached** (SATA controller)
- Do **not** change disk format

### 3.5 Network (Critical)

Configure **at least two adapters**:

| Adapter | Purpose | Setting |
|---------|---------|---------|
|---------|---------|---------|

|           |                   |                                     |
|-----------|-------------------|-------------------------------------|
| Adapter 1 | Management (mgmt) | Bridged Adapter <b>or</b> Host-Only |
|-----------|-------------------|-------------------------------------|

- Adapter Type: **Intel PRO/1000 MT Desktop**
- Promiscuous Mode: Allow All

Click **Finish** to import.

### Login Credentials

- Username: admin
- Password: admin

### Fix the kernel panic Issue (Step-by-Step – Proven)

#### Step 1: Power Off the VM

Ensure the VM is **completely powered off**.

#### Step 2: Remove Incorrect Disk Controllers

1. Open **VM Settings → Storage**
2. Remove:
  - IDE Controller
  - SCSI Controller (if present)
3. **Do NOT delete the disk file**

#### Step 3: Add Correct Controller (AHCI SATA)

1. Click **Add Controller**
2. Select **SATA Controller**
3. Set:
  - Name: SATA Controller
  - Type: **AHCI**
  - Ports: 2

#### Step 4: Attach Palo Alto Disk to SATA

1. Under SATA Controller → **Add Hard Disk**
2. Choose **Existing Disk**
3. Select the imported .vmdk
4. Ensure:
  - **Port 0**
  - **Device 0**

#### Step 5: Confirm Boot Order

##### System → Motherboard

- Boot Order:
  1. Hard Disk
  2. Optical
- Disable everything else

#### Step 6: Start the VM

Expected boot sequence:

- GRUB loads
- Kernel loads
- Root filesystem mounts
- Login prompt appears

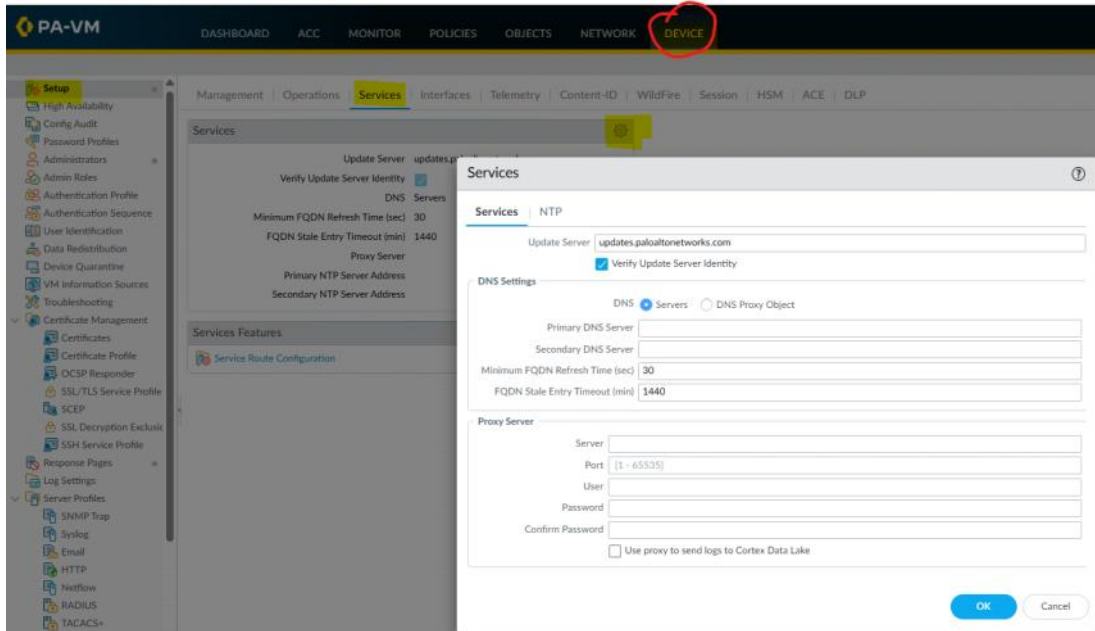
**Note - Palo Alto VM-Series on VirtualBox will only boot reliably if the root disk is attached to an AHCI SATA controller, Port 0, using the VMware OVA image.**

# Configuring DNS & NTP

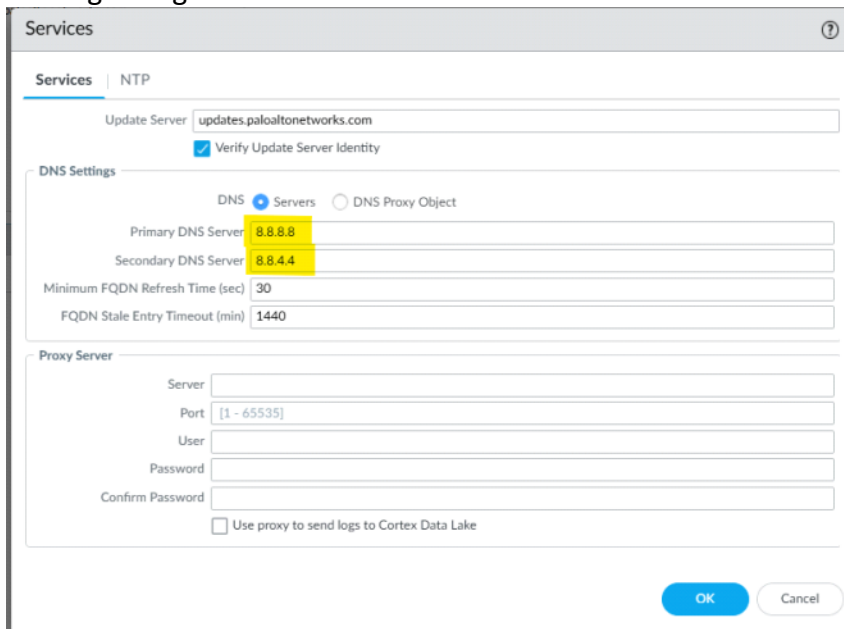
Thursday, December 25, 2025 9:09 PM

## Domain Name System (DNS)

- It helps in translating/resolving name (ex: [www.example.com](http://www.example.com)) into valid IP address (ex: 1.1.1.1)
- Device --> Setup --> Services --> gear icon:



Making changes:



## Network Time Protocol (NTP)

- It's a networking protocol used for synchronizing clock between computers over the network.
- Changing NTP config:

Services

Services

NTP

Primary NTP Server

NTP Server Address

Authentication Type

None

Secondary NTP Server

NTP Server Address

Authentication Type

None

OK

Cancel

Services

Update Server

updates.paloaltonetworks.com

Verify Update Server Identity

☒

DNS Servers

Primary DNS Server

8.8.8.8

Secondary DNS Server

8.8.4.4

Minimum FQDN Refresh Time (sec)

30

FQDN Stale Entry Timeout (min)

1440

Proxy Server

Primary NTP Server Address

in.pool.ntp.org

Primary NTP Server Authentication Type

None

Secondary NTP Server Address

Services Features

Service Route Configuration

Now commit the changes:

DASHBOARD

ACC

MONITOR

POLICIES

OBJECTS

NETWORK

DEVICE

Commit

Management

Operations

Services

Interfaces

Telemetry

Content-ID

WildFire

Session

HSM

ACE

DLP

Services

Update Server

updates.paloaltonetworks.com

Verify Update Server Identity

☒

DNS Servers

Primary DNS Server

8.8.8.8

Secondary DNS Server

8.8.4.4

Minimum FQDN Refresh Time (sec)

30

FQDN Stale Entry Timeout (min)

1440

Proxy Server

Primary NTP Server Address

in.pool.ntp.org

Primary NTP Server Authentication Type

None

Secondary NTP Server Address

Services Features

Service Route Configuration

Commit

Doing a commit will overwrite the running configuration with the commit scope.

Commit All Changes

Commit Changes Made By: admin

COMMIT SCOPE

LOCATION TYPE

device-and-network

Preview Changes

Change Summary

Validate Commit

Group By Location Type

Description

Commit

Cancel

Ping google using management network:  
# ping host google.com

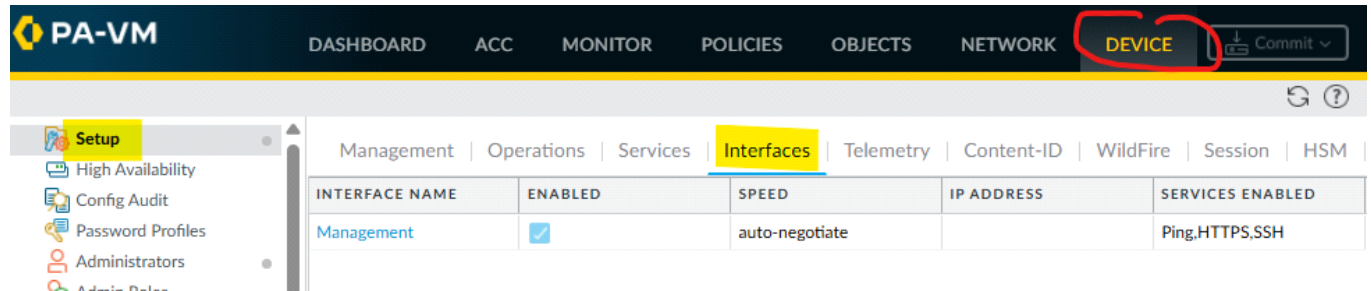




# Managing interfaces using PA-FW

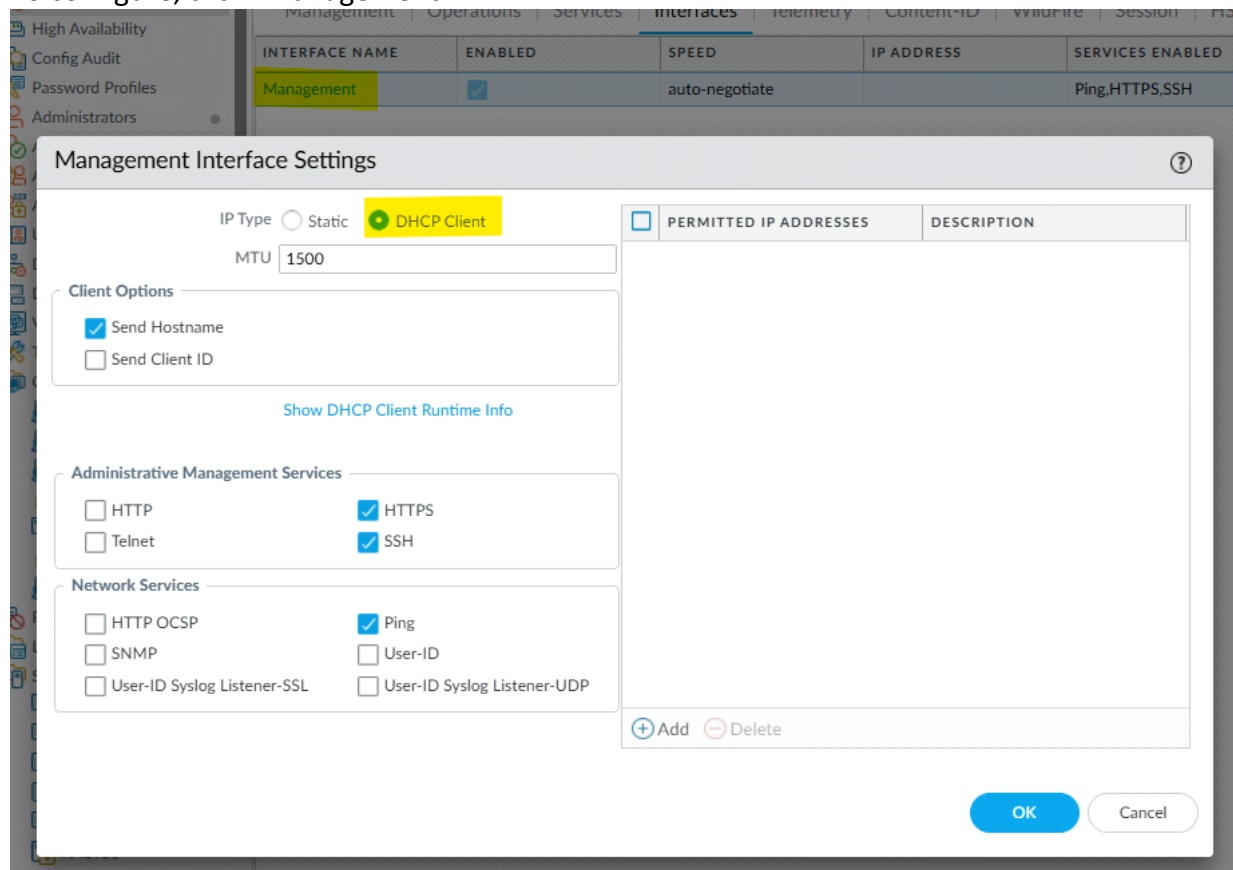
Thursday, December 25, 2025 10:30 PM

Go to interfaces:



- It allows ping, ssh and HTTPS connections. Rest all will be denied (as of now, but can be changed).

To configure, click "Management"



You can make changes here and then commit it.

To do the same using CLI:

```
admin@PA-VM> show system services

HTTP : Disabled
HTTPS : Enabled
Telnet : Disabled
SSH : Enabled
Ping : Enabled
SNMP : Disabled
```

Switch to config mode by typing "configure" :

*To remove the dhcp configuration:*

➤ # delete deviceconfig system type dhcp-client

*To add static configuration:*

➤ # set deviceconfig system type static

*To provide static IP address with subnet mask*

➤ # set deviceconfig system ip-address 192.168.1.100 netmask 255.255.255.0

*To finalize the configuration, we need to commit the changes.*

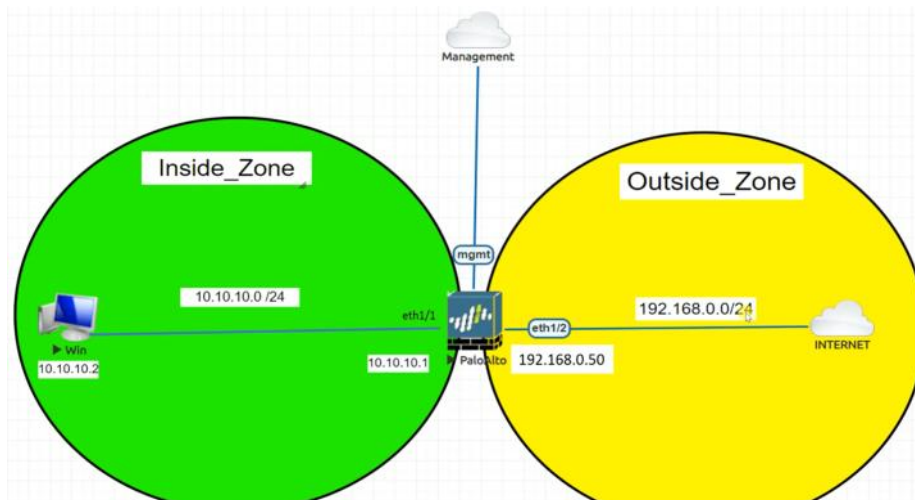
➤ # commit

## Basic requirements - firewall connectivity

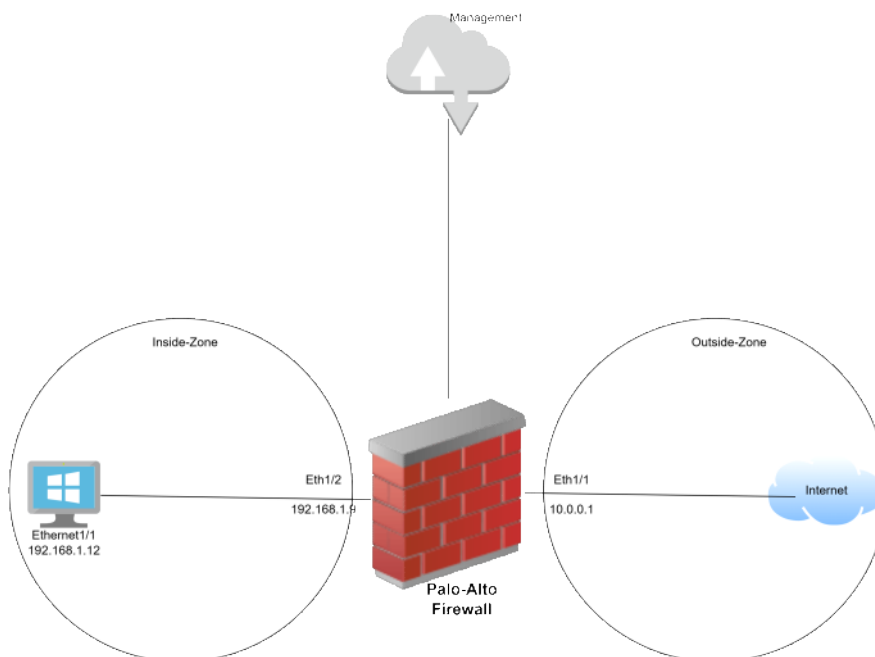
Thursday, December 25, 2025 10:59 PM

Below are the basic requirements to establish the connectivity through firewall:

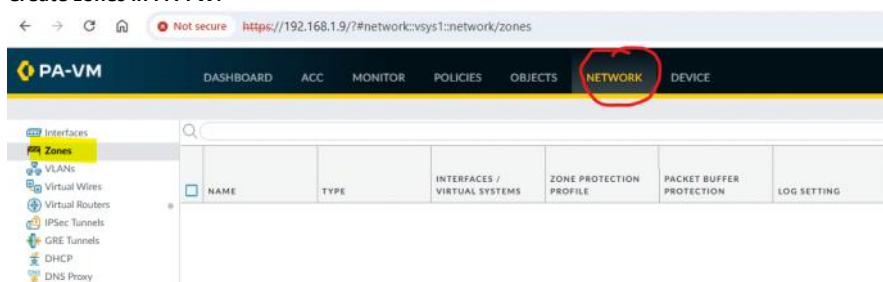
1. Define the firewall zones.
2. Configure the firewall interfaces
3. Configure routing to route the traffic between zones.
4. Define the NAT policy.
5. Define the security policy either to permit or deny.



My lab setup:



Create zones in PA-FW:



Then click on "Add" at the bottom to add zone(s).

Similarly, adding outside-zone.

| NAME         | TYPE   | INTERFACES / VIRTUAL SYSTEMS | ZONE PROTECTION PROFILE | PACKET B PROTECTI                   |
|--------------|--------|------------------------------|-------------------------|-------------------------------------|
| inside-zone  | layer3 |                              |                         | <input checked="" type="checkbox"/> |
| outside-zone | layer3 |                              |                         | <input checked="" type="checkbox"/> |

Go to Network -> Interfaces -> Ethernet -> Click on "ethernet1/1" and fill comment and int type:

Now click on IPv4 tab to config IP address:

Now to go to "Advanced" and create a management port for ping (not recommended for production):

### Ethernet Interface ?

Interface Name

Comment

Interface Type

Netflow Profile

Config | IPv4 | IPv6 | SD-WAN | **Advanced**

**Link Settings**

Link Speed  Link Duplex  Link State

**Other Info** | ARP Entries | ND Entries | NDP Proxy | LLDP | DDNS

Management Profile

MTU

☐ Adjust TCP MSS

IPv4 MSS Adjustment

IPv6 MSS Adjustment

☐ Untagged Subinterface

**OK** **Cancel**

Now configuring the "Ethernet 1/2" for internet:

### Ethernet Interface ?

Interface Name

Comment

Interface Type

Netflow Profile

**Config** | IPv4 | IPv6 | SD-WAN | Advanced

**Assign Interface To**

Virtual Router

Security Zone

**OK** **Cancel**

### Ethernet Interface ?

Interface Name

Comment

Interface Type

Netflow Profile

Config | IPv4 | IPv6 | SD-WAN | **Advanced**

**Link Settings**

Link Speed  Link Duplex  Link State

**Other Info** | ARP Entries | ND Entries | NDP Proxy | LLDP | DDNS

Management Profile

MTU

☐ Adjust TCP MSS

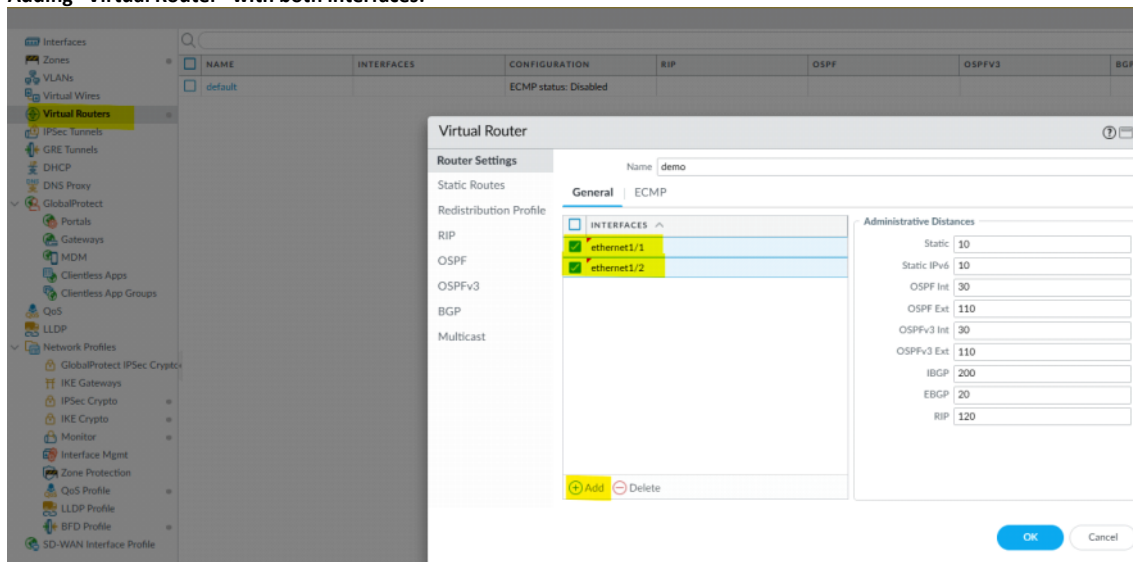
IPv4 MSS Adjustment

IPv6 MSS Adjustment

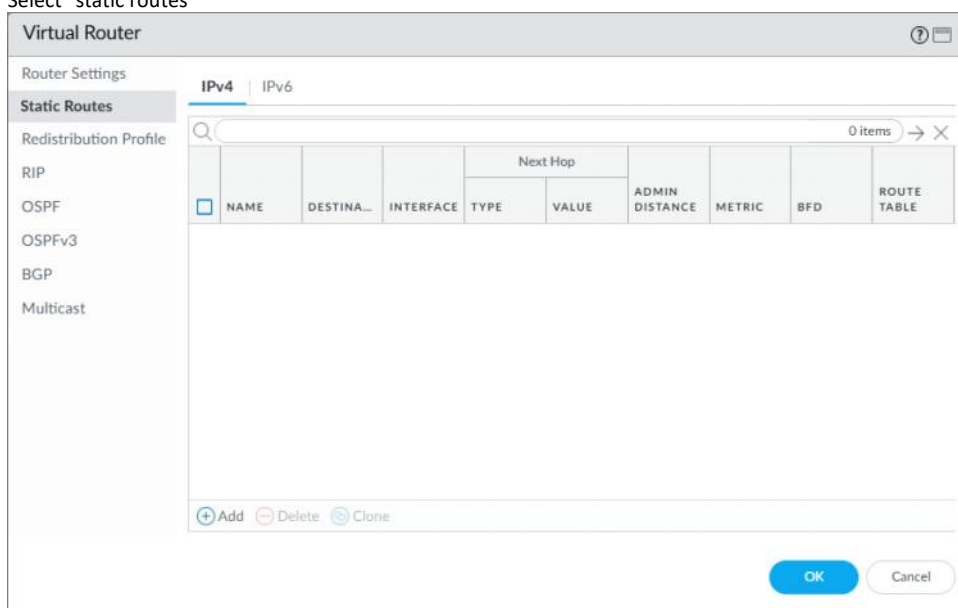
☐ Untagged Subinterface

**OK** **Cancel**

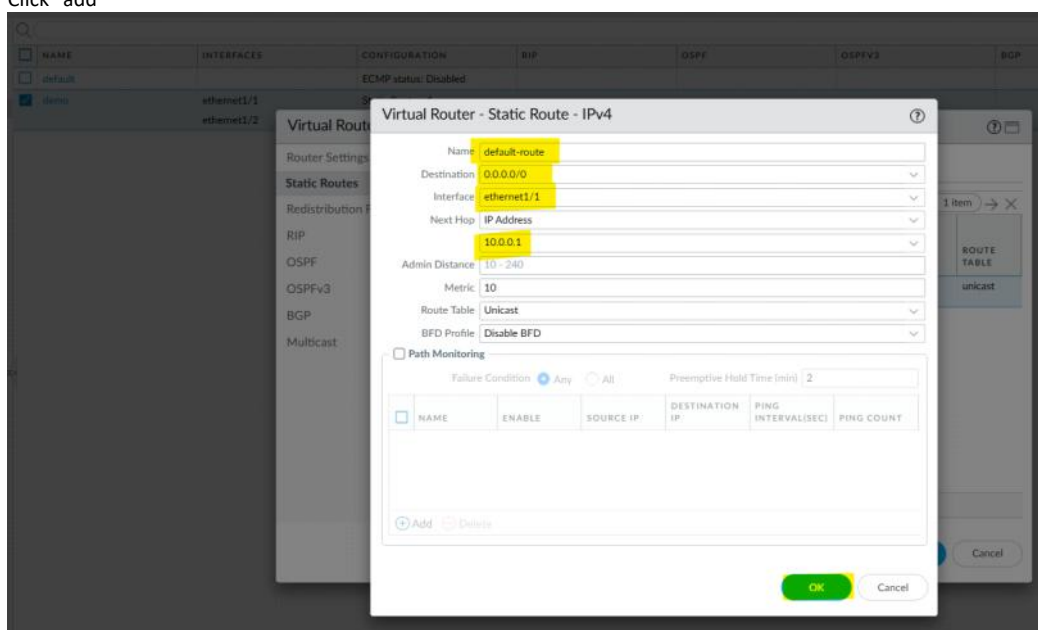
Adding "Virtual Router" with both interfaces:



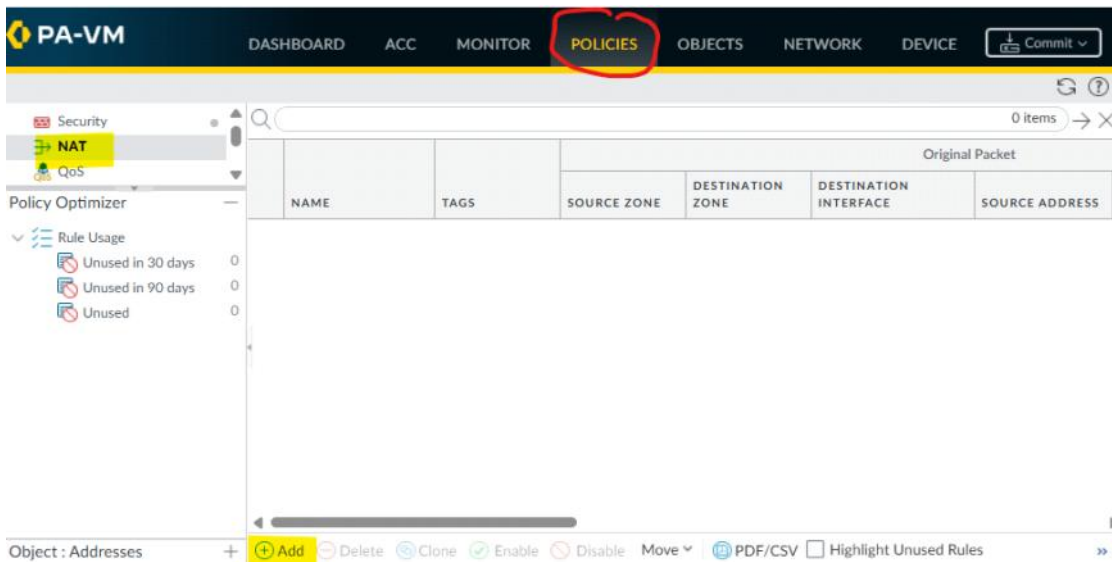
Select "static routes"



Click "add"



Now configuring the NAT Policy:



Adding name & type:

Select "Original Packet"

Select "Translated Packet":



**NAT Policy Rule** ?

General | Original Packet | **Translated Packet**

**Source Address Translation**

Translation Type: **Dynamic IP And Port**

Address Type: **Interface Address**

Interface: **ethernet1/2**

IP Address: **10.0.0.1/8**

**Destination Address Translation**

Translation Type: **None**

OK
Cancel

**PA-VM** DASHBOARD ACC MONITOR **POLICIES** OBJECTS NETWORK DEVICE Commit

---

Security

**NAT**

QoS

Policy Based Forwarding

Decryption

Tunnel Inspection

Application Override

Authentication

DoS Protection

SD-WAN

|   |            | Original Packet |             |                  |                       |                |
|---|------------|-----------------|-------------|------------------|-----------------------|----------------|
|   | NAME       | TAGS            | SOURCE ZONE | DESTINATION ZONE | DESTINATION INTERFACE | SOURCE ADDRESS |
| 1 | in2out-nat | none            | inside-zone | outside-zone     | any                   | any            |

Now configuring "Security Policy":

**PA-VM** DASHBOARD ACC MONITOR **POLICIES** OBJECTS NETWORK DEVICE

Security

**NAT**

QoS

Policy Based Forwarding

Decryption

Tunnel Inspection

Application Override

Authentication

DoS Protection

SD-WAN

|   |                   | Source |           |      |         | Destination |        |             |         |        |             |         |
|---|-------------------|--------|-----------|------|---------|-------------|--------|-------------|---------|--------|-------------|---------|
|   | NAME              | TAGS   | TYPE      | ZONE | ADDRESS | USER        | DEVICE | ZONE        | ADDRESS | DEVICE | APPLICATION | SERVICE |
| 1 | intrazone-default | none   | Intrazone | any  | any     | any         | any    | (intrazone) | any     | any    | any         | any     |
| 2 | interzone-default | none   | Interzone | any  | any     | any         | any    | any         | any     | any    | any         | any     |

**Security Policy Rule** ?

General | **Source** | Destination | Application | Service/URL Category | Actions

Name: **in2out-sec-policy**

Rule Type: **universal (default)**

Description:

Tags:

Group Rules By Tag: **None**

Audit Comment:

OK
Cancel

Select inside-zone for source:

**Security Policy Rule**

General | **Source** | Destination | Application | Service/URL Category | Actions

|                                                              |                                                              |                                                              |                                                              |
|--------------------------------------------------------------|--------------------------------------------------------------|--------------------------------------------------------------|--------------------------------------------------------------|
| <input type="checkbox"/> Any                                 | <input checked="" type="checkbox"/> Any                      | any                                                          | any                                                          |
| <input type="checkbox"/> SOURCE ZONE ^                       | <input type="checkbox"/> SOURCE ADDRESS ^                    | <input type="checkbox"/> SOURCE USER ^                       | <input type="checkbox"/> SOURCE DEVICE ^                     |
| <input checked="" type="checkbox"/> inside-zone              |                                                              |                                                              |                                                              |
| <input type="checkbox"/> Add <input type="checkbox"/> Delete | <input type="checkbox"/> Add <input type="checkbox"/> Delete | <input type="checkbox"/> Add <input type="checkbox"/> Delete | <input type="checkbox"/> Add <input type="checkbox"/> Delete |

☐ Negate

OK Cancel

Go to destination:

**Security Policy Rule**

General | Source | **Destination** | Application | Service/URL Category | Actions

|                                                              |                                                              |                                                              |
|--------------------------------------------------------------|--------------------------------------------------------------|--------------------------------------------------------------|
| select                                                       | <input checked="" type="checkbox"/> Any                      | any                                                          |
| <input type="checkbox"/> DESTINATION ZONE ^                  | <input type="checkbox"/> DESTINATION ADDRESS ^               | <input type="checkbox"/> DESTINATION DEVICE ^                |
| <input checked="" type="checkbox"/> outside-zone             |                                                              |                                                              |
| <input type="checkbox"/> Add <input type="checkbox"/> Delete | <input type="checkbox"/> Add <input type="checkbox"/> Delete | <input type="checkbox"/> Add <input type="checkbox"/> Delete |

☐ Negate

OK Cancel

Leave Application and Service/URL category as it is. Jump to "Actions" and

**Security Policy Rule**

General | Source | Destination | Application | Service/URL Category | **Actions**

**Action Setting**

Action:

☐ Send ICMP Unreachable

**Profile Setting**

Profile Type:

**Log Setting**

☒ Log at Session Start

☒ Log at Session End

Log Forwarding:

**Other Settings**

Schedule:

QoS Marking:

☐ Disable Server Response Inspection

OK Cancel

And validate:

|   | NAME              | TAGS | TYPE      | ZONE        | ADDRESS | USER | DEVICE | ZONE         | ADDRESS |
|---|-------------------|------|-----------|-------------|---------|------|--------|--------------|---------|
| 1 | in2out-sec-policy | none | universal | inside-zone | any     | any  | any    | outside-zone | any     |
| 2 | intrazone-default | none | intrazone | any         | any     | any  | any    | (intrazone)  | any     |
| 3 | interzone-default | none | interzone | any         | any     | any  | any    | any          | any     |

COMMIT the changes and you are done.

To validate the network is configured from PA-FW properly:

```
ping source 10.10.10.1 host 8.8.8.8
```

- This must ping, meaning everything is good.